Semi-supervised learning review (with code!) from methodology to application in real life

Axe: bridge gap between method and applications :rappelle des méthodes classiques de SSL deep et montrer leur connexion avec les méthodes utilisées en pratiques sur des datasets réels (qu'est ce qu'il faut rajouter): en segmentation par exemple. + parler du Sampler +SemiPy

Validation + PPI

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ACM Reference Format:
. 2018. Semi-supervised learning review (with code!) from methodology to application in real life. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 47 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

- Debut du SSL & importantes applications
- revue des revues + démontrer la nouveauté dans la notre
- Justifier choix de s'intéresser à la segmentation.

In many real-world scenarios, it is easy to collect large amounts of unlabelled data but costly and time-consuming to obtain corresponding labels, especially in tasks like medical image segmentation, where trained experts must manually annotate images (contour the organs, tissues or lesions possibly on 3D images), while acquiring the images is in the standard of care, depending on the pathology of the patients. In such cases, one may seek to leverage the unlabelled data to improve supervised learning performance, leading to the semi-supervised learning (SSL) framework, which sits between supervised and unsupervised learning. Regardless of the scenario or prior knowledge, the core promise of SSL is to improve performance by incorporating unlabelled data into a supervised learning framework, or to match performance levels using fewer labelled examples, thus reducing the cost and effort of annotation.

While most SSL methods have focused on classification, there has also been progress in regression settings [111]. A short review on recent regression methods is given in Appendix A. Several surveys have reviewed SSL methods in both traditional machine learning [29, 200, 276] and deep learning [164, 222] contexts. Particularly noteworthy is the application of SSL to image segmentation, and medical image segmentation in particular [40]. These tasks are compelling testbeds for SSL because they represent real-world situations where labelled data is expensive and scarce, but high performance is critical. In this case, SSL is not only advantageous but arguably necessary. Finally, we also propose the first literature review of semi-supervised validation.

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Manuscript submitted to ACM

Our semi-supervised learning vocabulary:

- Complete case: supervised learning baseline; model trained using only the labelled samples.
- Epoch: The definition of epoch is tricky in SSL. We advocate for a full pass on the labelled subset. Other works call an epoch as a full pass on the unlabelled subset [204].
- Classic assumptions: assumptions on the data distribution, on features and labels, used to justify the methods' performance. We cite the smoothness, cluster, low-density and manifold assumptions; See section 2.2.2.
- MCAR, MNAR MAR: classical assumption from the missing data literature characterising the missingness mechanism. (see Section 2.3.1).
- Safe SSL: SSL method with theoretical guarantees that are similar or stronger to the complete case baseline.
- Open-world SSL: unlabelled data might be drawn from classes that are unobserved.
- Class-imbalanced: data has unequal class representation.
- Long-tailed SSL: class frequencies follow a long-tailed distribution across labelled and/or unlabelled data
- Realistic SSL: unlabelled data includes out-of-distribution or class-unbalanced samples.
- Pseudo-label: a prediction of a teacher model used to train a student model (the teacher and the student can be the same model).
- Self-training: model generates and uses its own pseudo-labels to train on unlabelled data.
- Confirmation bias: Tendency of a model to reinforce its own errors during training.

1) On peut faire un tableau de notation ? 2) "classer le tableau vocabulaire": mettre que les notions qu'on explique vraiment après + faire des paquets, genre "hypothèses", "baselines", "implementation"...

2 WHAT IS SEMI-SUPERVISED LEARNING?

2.1 Brief notations

Formally, SSL assumes access to a labelled dataset $\mathcal{L} = ((x_1, y_1), ..., (x_{n_l}, y_{n_l}))$ drawn identically and independently from a distribution $p_{\mathcal{L}}(x, y)$, with the x and labels y, defined over $\mathcal{X} \times \mathcal{Y}$ and a unlabelled dataset $\mathcal{U} = (x_1, ..., x_{n_u})$ drawn identically and independently from the marginal distribution $p_{\mathcal{U}}(x)$. Most SSL methods assume that the marginal distribution in the labelled and unlabeled dataset is the same, implying $p_{\mathcal{U}}(x) = p_{\mathcal{L}}(x)$. In the following, this common distribution is denoted as p, and we will introduce more rigorous notations in Section 2.3.1. Usually, n_l is assumed to be small and $\mathcal{U}$ larger than n_u. In the limit of $n_u = 0$, we fall back into the classic supervised learning framework, and conversely, if $n_l = 0$, it is unsupervised learning.

SSL methods primarily aim to learn a model of the distribution of labels given the features (i.e., the conditional $p(y|x)$). Usually, we approximate this distribution using a set of models f_θ, parametrised by $\theta \in \Theta$. In general, for classification with neural networks, we can map $\mathcal{Y}$ to $\{1, \ldots, C\}$ with C the number of classes, and define

$$p_\theta(.|x) = \text{SoftMax}(f_\theta(x)) = \left(\frac{\exp f_\theta(x)_i}{\sum_{j=1}^{C} f_\theta(x)_j} \right)_{i=1}^{C}.$$

2.2 First intuitions on semi-supervised learning

2.2.1 When can it help? Intuitively, SSL exhibits parallels with human concept acquisition. For instance, a learner may be presented with a single labelled instance of a small four-legged animal identified as a cat. Subsequent encounters with numerous unlabeled instances of similar animals, varying in colour, shape, and other attributes, allow the learner

Table 1. Notations table.

$\mathcal{L}/\mathcal{U}$	Labelled/Unlabelled dataset
n_l/n_u	Number of labelled/unlabelled samples
$p, p_{\mathcal{L}}, p_{\mathcal{U}}$	True distributions
$\theta \in \Theta$	model's parameters in admissible parameters
f_θ/p_θ	model's outputs/output probabilities
$\hat{\theta}$	stop gradients' copy of θ

to progressively refine their internal representation of the category "cat" [276, Chapter 7]. In this way, humans naturally leverage unlabelled data to enhance their learning, so why shouldn't machines do the same?

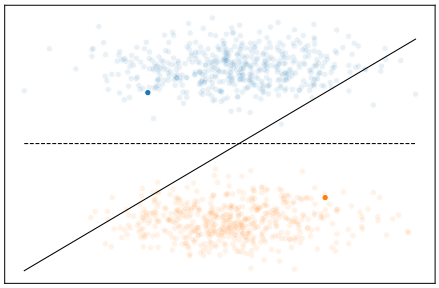

Fig. 1. Intuition behind semi-supervised learning. The plot illustrates a binary classification task with both labelled and unlabelled data. Samples are drawn from two distinct 2D Gaussian distributions ("blobs"). The two darker points represent the only labelled examples, while the lighter points are unlabelled and coloured according to their true (but unobserved) class labels. The solid line indicates the decision boundary learned by a supervised classifier trained solely on the labelled data. The dashed line represents the optimal decision boundary that could be obtained by leveraging the full dataset, including the unlabelled samples.

Figure 1 provides an intuitive illustration of how unlabelled data can improve a binary classifier. In this example, 1,000 samples are drawn from two 2D Gaussian blobs, with only two randomly labelled (shown as darker points). A classifier trained solely on these labels is likely to place the decision boundary perpendicular to the segment connecting them. In contrast, incorporating the unlabelled data enables recovery of the optimal boundary, horizontal and aligned with the low-density region between the blobs.

The key insight behind SSL is that unlabelled data can reveal the structure of the marginal distribution $p(x)$, which in turn helps to better estimate the conditional distribution $p(y|x)$. By exploiting this structure, SSL methods can significantly improve performance over purely supervised approaches, especially when labelled data is scarce.

On the other hand, SSL systematically fails to improve over a supervised learning baseline when knowledge of the marginal distribution $p(x)$ provides no useful information about the conditional distribution $p(y|x)$. For example, if features x are strict causes of the target y (causal relationship), then unlabelled data, informing only about $p(x)$, cannot help in the prediction [197]. However, when some features are effects of the target, their distribution conditioned on causal features can help refine predictions [114]. For instance, in a medical diagnosis setting, causal factors (age, genetics, etc.) and effect features (symptoms, imaging, etc.) interact in this way, enabling unlabelled data to enhance learning through anticausal links. Prior knowledge of causal structure thus plays a key role in determining when SSL is effective.

Seeger [200] argues that generative models are naturally well-suited to SSL because they explicitly model the joint distribution $p(x, y)$, and thereby inherently capture the marginal distribution $p(x)$. In contrast, classical discriminative models focus solely on modelling $p(y|x)$ (or directly, on mapping data points x to labels y) and therefore cannot directly exploit unlabelled data, since their likelihood-based objective relies only on labelled inputs. To make SSL possible with discriminative models, one must introduce some form of parameter-sharing between the conditional model and the marginal distribution $p_\theta(x)$ during the learning process; this dependency is crucial for extracting information from unlabelled data [200, Section 1.3.2]. From this perspective, SSL can be seen as supervised learning augmented with additional information about the data distribution provided by unlabelled samples [200]. However, this augmentation relies heavily on prior assumptions, for example, that labelled and unlabelled data are drawn from the same distribution or that specific structural properties hold in the data. Such reliance makes SSL more vulnerable than purely supervised approaches: when labels are scarce or unevenly distributed, poorly matched or misleading unlabelled data can easily harm the learning process or degrade performance rather than improve it.

2.2.2 Assumptions of semi-supervised learning. SSL is effective only when the marginal distribution of features $p(x)$ contains information about the conditional distribution $p(y|x)$. In such cases, unlabelled data can improve our knowledge of $p(x)$ and therefore, benefit the prediction of $p(y|x)$. To enforce this dependency, several standard assumptions on the data distribution have been proposed [29, 222]:

- **Smoothness assumption**: If two samples x_1, x_2 are close in a high-density region, their outputs y_1, y_2 should also be similar [29, Section 1.2.1]. This implies that the prediction function should vary smoothly in dense regions and applies to classification, regression, and segmentation. For instance, in perturbation-based SSL, admissible transformations $\mathcal{T}$ are assumed not to change the underlying label, $p(y|x) = p(y|T(x))$ with $T \in \mathcal{T}$. This assumption underpins consistency regularisation, which encourages stable predictions across perturbed views of the same input (see Section 4.1.1).
- **Low-density assumption**: Decision boundaries should lie in low-density regions [164, 222], a principle particularly relevant for classification.
- **Manifold assumption**: Data are assumed to approximately lie on a low-dimensional manifold embedded in a high-dimensional space [29, Section 1.2.3], mitigating the curse of dimensionality. Stronger forms posit that each class corresponds to a separate manifold [13, 222], as Wasserman and Lafferty [233] show that the manifold assumption alone is insufficient for SSL. Mey and Loog [149] further highlight ambiguities between formulations.
- **Cluster assumption**: Samples within the same cluster, defined as a group of similar samples, are likely to share the same label [29, Section 1.2.2]. Unlike the manifold assumption, this does not imply a single cluster per class but rather that labels are consistent within each cluster.

Figure 2 illustrates the two-moon dataset under different levels of noise. On the left, all assumptions are satisfied; on the right, the cluster and low-density assumptions fail, the manifold assumption holds only in the weaker sense described by Chapelle et al. [29]. The smoothness assumption is ambiguous: nearby points generally share the same label, but the assumption breaks down near the decision boundary.

These assumptions are closely related in classification. The cluster, low-density, and manifold assumptions are often seen as variations of the same principle [29, 222]: nearby samples share labels (cluster), boundaries avoid dense regions (low-density), and label consistency holds on low-dimensional structures (manifold). The cluster assumption can also be viewed as a special case of smoothness or manifold assumptions [13]. Despite this overlap, the distinctions between

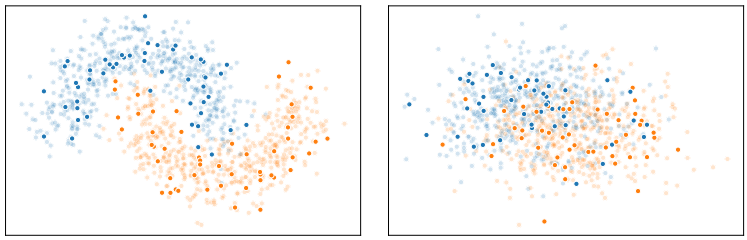

Fig. 2. Two-moon dataset with different variances. Lighter points are unlabelled; darker points are labelled. **(Left)** All assumptions hold. **(Right)** The manifold assumption holds in the weak sense of Chapelle et al. [29], the smoothness assumption is uncertain, and the cluster and low-density assumptions do not hold.

these assumptions help us understand and design SSL algorithms. Schölkopf et al. [197] argue that these assumptions essentially enforce a causal direction from x to y, aligning with the conditions for SSL to be effective. **Importantly, these assumptions cannot be tested directly and must be inferred from prior knowledge of the data.** This limitation underscores the risks of applying SSL without careful consideration, especially in real-world applications where misleading, unlabelled data can harm performance. For a detailed discussion of such failure cases, see Mey and Loog [149].

2.2.3 Earliest works on semi-supervised learning. To the best of our knowledge, the origins of SSL trace back to the 1960s, with the introduction of self-learning methods, also referred to as self-labelling or pseudo-labelling, by Scudder [198]. In modern deep SSL literature, this method is widely known as pseudo-labelling (PL). For clarity, we refer to the deep learning variant as PL, and use self-training to denote the more general, model-agnostic version. Self-training is a simple yet powerful wrapper algorithm compatible with any supervised learning model. The procedure involves training a model on labelled data, then iteratively augmenting the training set by assigning *pseudo-labels* to unlabelled data using the model's predictions. These pseudo-labels may be filtered based on confidence or uncertainty measures. The idea was introduced in several nearly contemporaneous works in the sixties, such as Scudder [198], Fralick [72], and Agrawala [1]. **Notably, many of today's most prominent deep discriminative SSL approaches build upon this foundational concept.**

Shortly thereafter, several works considered the use of additional unlabelled data to compute the linear discriminant rule with Gaussian and homoscedasticity assumption on the conditional densities [93, 163]. Similar methods under multinomial assumptions were examined by Cooper and Freeman [47]. Around the same period, Vapnik 223, Remark 7.11.2 introduced the transductive learning framework, which focuses on predicting only for the unlabelled test set without the goal of generalising beyond it.

Interest in SSL surged in the 1990s, particularly in natural language processing applications [19, 45, 102, 157, 249]. Although the methodological foundations of SSL were laid in the 1960s, the term semi-supervised learning did not emerge until the late 1980s [20], and remained sporadic throughout the 1990s [14, 148]. It was only in the early 2000s that the term gained widespread adoption, especially following the influential work of Chapelle et al. [29].

2.3 Our vision of SSL

The preceding section examined the principal assumptions underlying SSL. We argue that a more rigorous theoretical framework is required to precisely characterise the conditions under which SSL methods can be expected to yield

improvements. In what follows, we first develop a formal perspective on SSL as a problem of learning with missing data (missing labels), an intrinsic yet comparatively underexplored aspect of the paradigm. Secondly, we propose recommendations for properly evaluating SSL methods.

2.3.1 Labels as missing data. The SSL framework can naturally be interpreted through the lens of missing data theory [133], as the unlabelled data are observations with missing labels. This perspective has already been adopted in past works [2, 196, 200], and we choose to highlight it in what follows. In this framework, we consider n samples $((x_1, y_1), \ldots, (x_n, y_n))$ drawn from the distribution $p(x, y) = p(y|x)p(x)$, but some of the labels are missing. A binary random variable r is introduced to indicate whether a data point is labelled ($r = 1$) or not ($r = 0$). This allows us to avoid modelling two separate distributions for the labelled dataset and the unlabelled dataset, $p_{\mathcal{L}}(x, y)$ and $p_{\mathcal{U}}(x)$, as this notation naturally induces:

- A labelled set $\mathcal{L}$ of n_l samples drawn from $p(x, y|r = 1)$, and
- An unlabelled set $\mathcal{U}$ of n_u samples from $p(x, y|r = 0)$.

Another advantage of adopting this view on missing data is that it allows us to refer to the missing-data mechanisms when discussing the classical assumptions on the distribution, and to consider their possible relaxation in order to better reflect real-world scenarios. The missing-data mechanism is the conditional distribution of the missing-data indicator r given the variables x, y. Rubin [189] classifies the missing data into three types, depending on the cause of missingness. This typology is directly applicable to SSL.

Missing Completely At Random (MCAR). The labels are said MCAR, when the missingness of a label y is independent of its features x and the value of the label itself [153, 204, 217, 240],

$$r \perp\!\!\!\perp x, y,$$

meaning $p(r = 1|x, y) = p(r = 1) = \pi = 1 - p(r = 0)$ for some constant $\pi \in (0, 1)$. Under MCAR, the labelled and unlabelled data distributions are identical: $p(x, y|r = 0) = p(x, y|r = 1)$. As previously mentioned, theoretical works on SSL widely assume this assumption [32, 146, 147, 149, 163, 264], while most of the popular empirical methods assume it without stating it.

Missing At Random (MAR). In real-world situations, the missingness of a label may depend on the features x but not on the labels y. In the context of medical image segmentation, it can happen when data are collected using two different imaging devices, but only one device's outputs are annotated, the one with the images of the best quality. The labels are then said MAR if:

$$r \perp\!\!\!\perp y|x,$$

meaning $p(y|r = 0, x) = p(y|r = 1, x)$. Here, the features can follow different marginal distributions in the unlabelled dataset and in the labelled dataset: $p(x|r = 0) \neq p(x|r = 1)$.

Missing Not At Random (MNAR). Both of these missingness types does not model what Sportisse et al. [207] call "unbalanced class popularity", when people does not label classes in equal proportions regardless to their nature or to the quality of the data. When the missingness of a label can depend on both the values of the label itself and of the features, the labels are said to be MNAR. There are specific subtypes relevant to SSL:

- Self-masked MNAR: the missingness of a label depends solely on the label itself and we assume:

$$r \perp\!\!\!\perp x|y,$$

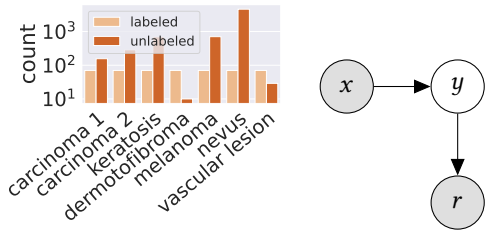

Fig. 3. Left panel: MNAR labels for dermaMNIST [44] (log count of labelled and unlabeled images per class). Right panel: Structural causal graph of the self-masked mechanism. The nodes in grey represent fully observed variables, and the edge from x to y means that x causes y.

meaning that $p(x|r = 1, y) = p(x|r = 0, y)$, that is, conditionally to the class y, the features should have the same distribution in both the labelled and unlabelled datasets. However, the features may follow different marginal distributions in the unlabelled dataset and in the labelled dataset, as is also the case with MAR labels. This type of missingness also allows modelling unbalanced class popularity. To illustrate this, we use the same example as in Sportisse et al. [207], describing a realistic self-masked MNAR scenario in which a medical doctor aims to classify medical conditions equally by selecting 70 images per class for labelling. This unbalanced class popularity is represented in the left panel of Figure 3, while the right panel shows the causal relationships between the variables x, y and the missing-data indicator r.

- Unseen class MNAR: Some unlabelled samples may belong to entirely unseen classes [88], rendering them unannotatable. Let us denote $\mathcal{Y}_u$ the set of missing labels in the unlabelled dataset. When $\mathcal{Y}_u \setminus \mathcal{Y} \neq \emptyset$, there is at least a new class in the unlabelled dataset, unseen in the labelled dataset. This implies label distribution mismatch, i.e. $p(y|r = 1) \neq p(y|r = 0)$ and then $r \not\perp\!\!\!\perp y$.

For MNAR labels, the missing-data mechanism is not ignorable in likelihood-based inference [2], and even the supervised baseline becomes biased, unless the mechanism is taken into account [2, 147, 207].

In the following, adopting the missing data literature vocabulary, we will also refer to the supervised baseline (model trained using only samples from $\mathcal{L}$) by the term *complete case* [189], as mentioned in Table **??**.

rajouter un schéma comme le left panel pour le MCAR/MAR de balanced popularity ?

2.3.2 On the risk of SSL:. Despite its potential, SSL also presents important risks. The inclusion of unlabelled data can sometimes degrade performance rather than improve it [222]. This is especially problematic when labelled data is limited, as it hampers the ability to reliably assess generalisation [160]. In fact, SSL only outperforms supervised learning baselines (equivalent to the complete case approach in the missing data literature Rubin 189) under specific conditions [132, 197, 202]. This risk of degradation has been particularly examined for generative models in Chapelle et al. [29, Chapter 4].

Empirical studies reveal significant limitations of current SSL methods, even in the MCAR setting. For instance, Zenk et al. [257] showed that FixMatch [204], despite its state-of-the-art status, yields negligible improvements when subjected to realistic evaluation in medical imaging, raising doubts about its practical utility. Moreover, Zhu et al. [279] demonstrated that pseudo-labelling can introduce severe biases across population sub-classes, in some cases driving class-level accuracy to zero [33]. These findings underscore that SSL, far from being universally beneficial, may exacerbate disparities or even undermine performance when applied unquestioningly. In fact, standard benchmarks can

overestimate their effectiveness [230], while real-world datasets, particularly in sensitive domains such as medicine, expose their fragility [257]. These factors highlight that, without careful consideration of the data distribution and methodological assumptions, SSL may not only fail to deliver benefits but also undermine model reliability and fairness.

To mitigate this risk, previous works have introduced the concept of *safe* semi-supervised learning, methods which never reduce predictive performance by introducing unlabelled data [88, 132]. In this work, we propose a loose definition of safeness as follows: *an SSL algorithm is safe if it has theoretical guarantees that are similar or stronger to the complete case baseline*. The "theoretical" part of the definition is motivated by the fact that any empirical assessment of generalisation performances of an SSL algorithm is jeopardised by the scarcity of labels. "Similar or stronger guarantees" can be understood in a broad sense since there are many kinds of theoretical guarantees (e.g. two methods may be both consistent, have similar generalisation bounds, and be both asymptotically normal with related asymptotic variances).

Unfortunately, most popular deep SSL methods lack such guarantees unless strong, typically untestable assumptions about the data distribution are made [149]. We discussed these assumptions in detail in Section 2.2.2).

2.3.3 Evaluation of SSL methods. In purely supervised learning, evaluation typically involves the choice of a benchmark dataset, data partitioning (into training, validation, and test sets), and hyperparameter tuning. Additional challenges arise when there are missing labels, including the need to adjust additional hyperparameters related to the SSL method itself, and data partitioning with few labelled data points. Moreover, SSL methods are most often evaluated on fully labelled benchmark datasets (see examples in the paragraph below), which are then artificially adapted to the semi-supervised setting by randomly selecting a subset of labelled and unlabelled data. The main difficulty, since the missing labels are artificial, is to consider realistic scenarios.

Benchmark datasets: Historically, SSL methods were evaluated on tabular UCI datasets [8], as discussed in Chapelle et al. [29, Chapter 21]. More recent deep learning-based SSL methods are commonly benchmarked on computer vision datasets such as CIFAR-10 and CIFAR-100 [113], SVHN [155], and ImageNet [190], as well as, less frequently, on NLP datasets like IMDb [142] and Yelp Reviews [266]. Recent benchmarks extended this list of datasets to Audio datasets [230]. MedMNIST datasets [246] with biomedical images have also been considered, taking advantage of the medical context that is conducive to more realistic SSL scenarios with MNAR labels [196, 207]. Finally, the STL-10 dataset [43] is a subset of ImageNet with 10 classes with only a few labels, specifically designed for evaluating SSL methods beyond the MCAR case.

Guidelines for more realistic evaluation. Oliver et al. [160] were among the first to critically assess the gap between SSL research and real-world applicability. Despite receiving substantial attention, the evaluation standards for SSL methods have not significantly evolved since then. Here, we summarise four key guidelines we consider essential for realistic SSL evaluation:

(1) **A shared implementation framework**: To ensure comparability, SSL methods should be implemented within a common framework using standardised network architectures, training procedures, and evaluation protocols. While many methods adopt similar architectures (e.g., 13-layer CNNs or WideResNets), differences in implementation details (optimisers, training steps, learning rate schedules, pre-training) lead to significant variability in results. The importance of reproducibility is especially heightened in sensitive domains like medical imaging, as discussed by Colliot et al. [46]. Moreover, many modern SSL algorithms are composed of multiple components (e.g., Virtual Adversarial Training and entropy minimisation in 153), making it important to understand the contribution of each element, typically through ablation studies.

(2) **High-quality complete case**: SSL methods must be compared against a strong supervised baseline to quantify the added value of unlabelled data, which is not systematic as we have seen in Section 2.3.2. The most appropriate approach is to use the same model architecture while removing the unsupervised component (e.g., excluding the unsupervised loss in discriminative models). Surprisingly, this is not always respected in practice; for example, the influential FixMatch method [204] omits a supervised baseline entirely. Moreover, many SSL methods have several components

(3) **Diverse and realistic benchmark datasets**: Many SSL methods implicitly assume that the labels are MCAR. These methods are often evaluated on benchmark datasets only, where the ratio of labelled-to-unlabelled data is uniform across classes, and frequently only in class-balanced scenarios. However, real-world datasets rarely follow these ideal conditions. To ensure robustness and practical applicability, SSL methods should be evaluated under varying degrees of class imbalance and more realistic MAR or MNAR mechanisms. Additionally, performance should be assessed across different proportions of labelled-to-unlabelled data, to guarantee scalability.

(4) **Small validation sets**: Hyperparameter tuning should be performed using realistic validation sets, reflecting the scarcity of labelled data that motivates SSL in the first place. Alarmingly, several recent studies (e.g., 187, 230) use the test set for early stopping, which leads to optimistic and non-generalisable results (see Section 6).

Beyond these guidelines, we stress the importance of evaluating SSL methods on more diverse and realistic datasets. This includes testing under label imbalance (e.g., CIFAR-10-LT [53]) and extending beyond vision tasks to domains such as audio and text, as done in the USB benchmark [230]. A realistic validation protocol must be paired with an independent test set to avoid biased evaluation. Furthermore, because deep SSL methods often integrate multiple components, thorough ablation studies are essential to disentangle their individual contributions.

Finally, several benchmark studies have concluded that no SSL method consistently outperforms others across all datasets, a result aligned with the "no free lunch" theorem. Chapelle et al. [29, Chapter 21] compared eleven SSL algorithms and found that none uniformly outperformed the others or the complete case baseline across eight datasets. Similarly, Wang et al. [230] benchmarked deep discriminative SSL models across vision, NLP, and audio tasks, with the same conclusion: **SSL may improve performance, but not universally, and not always beyond a well-tuned supervised baseline.** Therefore, as the performance of SSL methods largely depends on how the dataset is constructed (e.g., label distribution, class balance), it becomes even more critical to clearly state the assumptions underlying each method.

2.4 Review purposes

SSL can be divided into inductive and transductive paradigms [223, Chapter 10, Section 9]. Transductive learning assumes access to a labelled training set and a fixed unlabelled test set, with the goal of predicting labels only for this test set. In contrast, inductive SSL learns a general prediction function over the input space $\mathcal{X}$, enabling predictions on unseen data. In this work, we classify inductive methods as deep versus non-deep, and generative versus discriminative. Yang et al. [248] recently reviewed generative SSL approaches. In this work, we instead focus on inductive deep discriminative methods, which currently dominate the field and are particularly relevant for medical image segmentation.

Our goal is to bridge the gap between methodological developments in SSL and their scalability to real-world applications. As a case study, we examine medical image segmentation, a domain with abundant unlabelled scans but limited expert annotations, which are costly and time-consuming to obtain. SSL offers an attractive solution, yet

```python
from sklearn import datasets
X, _ = datasets.make_blobs(centers=3, cluster_std=0.5, random_state=0)
# Take a nonparametric model and fit it using MI as objective
from gemclus.nonparametric import CategoricalModel
mi_model = CategoricalModel(n_clusters=3, gemini="mi")
y_pred = mi_model.fit_predict(X)
# Same nonparametric model, optimise a distance-regularised MI
gemini_model = CategoricalModel(gemini="mmd_ova", random_state=0,
    learning_rate=1e-2)
y_pred_regularised = gemini_model.fit_predict(X)
```

Fig. 4. Example of code (code de Louis)

naïvely extending existing SSL methods comes at a risk: models may propagate errors from unreliable pseudo-labels or produce overconfident yet clinically incorrect predictions. Moreover, in applications such as lesion detection or tumour delineation, these risks can directly impact diagnosis and treatment planning. Developing safe semi-supervised methods is therefore essential for clinical deployment (see Section 2.3.2).

Another key challenge lies in model validation. Given the scarcity of annotated data, large-scale evaluation with expert labels is often impractical. Nevertheless, reliable validation is critical to ensure generalisation for clinical use. This underscores the need for unsupervised or semi-supervised validation techniques that can assess performance with few or no labels. Such methods would enable continuous monitoring of segmentation quality, early detection of failure modes, and more efficient use of limited annotations, ultimately supporting the safe and trustworthy adoption of machine learning in healthcare.

In summary, this review first surveys inductive deep discriminative SSL methods for classification, then examines advances in safe learning and validation strategies, and finally discusses their extension to the segmentation of medical images.

Semipy introduction

3 FROM CLASSIC METHODS TO A GENERAL TAXONOMY OF SSL

Classical approaches laid the theoretical and methodological foundations of the field, shaping how unlabelled data can be leveraged to improve learning under limited supervision. In this section, we first provide an overview of the main families of non-deep SSL methods and then introduce our taxonomy of SSL, which builds on these foundations while offering a unifying vision of both non-deep and deep methods.

3.1 An atlas of classic (non-deep) methods

Self-training or pseudo-label methods: The pseudo-label (PL) method, introduced in the 1960s [1, 72, 198], is one of the earliest semi-supervised learning techniques and remains foundational today, especially in deep learning, where it underpins methods like FixMatch [204]. Also known historically as self-training, self-labelling, or decision-directed training, pseudo-labelling follows an iterative process of predicting labels for unlabelled data, selecting confident predictions, and retraining the model with both real and pseudo-labels. This process, often likened to an Expectation–Maximisation algorithm [29], benefits from simplicity and flexibility, although its effectiveness heavily depends on model choice, loss function, and pseudo-label quality. Self-training gained notable interest in NLP through Yarowsky's algorithm for word sense disambiguation [249], and remains a central strategy in modern SSL frameworks.

Co-training: Co-training extends self-training by using multiple classifiers that iteratively teach one another through pseudo-labels, originally introduced by Blum and Mitchell [19] using two views of web data (page text and anchor text). A key requirement is the diversity of prediction: models must make errors that are sufficiently different to benefit from each other [222, 274]. While classic co-training assumes natural multi-view data, alternative approaches generate artificial views via feature selection [37, 250], random projections [225], or PCA [265]. Diversity can also arise from using different models or hyperparameters [83, 272]. Co-training has been extended to tri-training with agreement rules across three classifiers [61, 273], and to co-forest methods leveraging random forests [126]. These models often act as ensembles, combining predictions to reduce classification error, and co-regularisation frameworks formalise this collaboration [201].

Boosting: Self-training and co-training can be naturally extended to boosting methods, where an ensemble of learners is trained sequentially with sample weights accentuating misclassified points. By introducing pseudo-labels for unlabelled data, boosting methods can be adapted to semi-supervised learning. For example, Grandvalet et al. [86] modified AdaBoost by incorporating an unsupervised loss on pseudo-labelled data, influencing sample weights across both labelled and unlabelled datasets (a technique later extended to gradient boosting [56]). ASSEMBLE [15] simplifies this by directly treating pseudo-labels as ground truth, making it model-agnostic. SemiBoost [145] further refines pseudo-label selection using a confidence score based on data similarity graphs. See van Engelen and Hoos [222] for a broader review of SSL boosting methods.

Maximum-margin methods: Maximum-margin methods aim to enforce the low-density separation assumption by pushing the decision boundary away from unlabelled data. This is typically done by adapting supervised margin-based models like support vector machines (SVMs) to incorporate unlabelled data, leading to semi-supervised SVMs (S3VMs) [223]. Though originally transductive, S3VMs can be seen as inductive since they define a decision function over the whole input space. However, the inclusion of unlabelled data introduces non-convexity by adding constraints that encourage boundaries to lie in low-density regions, regardless of class. Extensions like semi-supervised Gaussian processes [118] introduce a "null category" to repel the boundary from uncertain regions. Alternative approaches, such as entropy minimisation [85], encourage confident predictions on unlabelled data and are widely used in deep learning. Others use information-theoretic regularisation, penalising mutual information between features and predictions to discourage complex boundaries in dense regions [48, 112].

Manifold methods: Self-training and co-training rely on the smoothness assumption, while maximum-margin methods rely on low-density separation. The remaining key assumption in semi-supervised learning is the manifold assumption: data lie on low-dimensional manifolds where nearby points share labels. Two main strategies leverage this: manifold regularisation, which penalises differing predictions for nearby points on a graph, and manifold approximation, which seeks to estimate the underlying manifolds (via contrastive autoencoders [185], for instance). A notable example is Laplacian SVM [13], which combines standard SVM loss with a graph-based smoothness term over both labelled and unlabelled data. Extensions include twin SVMs [178] and deep learning adaptations [140, 236], where graphs are constructed from model predictions. However, graph construction can lead to overfitting, prompting ensemble methods to stabilise manifold regularisation [79].

Generative models: Generative models naturally incorporate assumptions about feature distributions, making them appealing for SSL. While widely explored in deep learning, traditional generative approaches like mixture models have seen limited SSL use. Cozman et al. [50] showed that unlabelled data can improve classification under correct model

assumptions, though these are rarely testable or valid in practice. Moreover, generative models typically underperform compared to discriminative ones [200]. Still, successful applications exist, such as Nigam et al. [158]'s EM-based multinomial mixture model for text classification. See 29, Chapter 4 and Part I for guidance and context.

Transductive methods: Many transductive methods are graph-based, building a similarity graph over the entire dataset and enforcing that similar samples receive similar labels. The predictor is trained to match true labels while encouraging smooth label transitions across the graph, often under manifold and smoothness assumptions. Early approaches like min-cut and Gaussian random fields propagate labels using graph connectivity, but suffer from scalability and binary-label constraints. Graph construction (how nodes are connected and how similarities are measured) significantly impacts performance, with methods ranging from ϵ-neighbourhoods and kNN to more sophisticated adaptive and sparsity-based techniques. Recent works explore deep transductive learning by integrating label propagation with neural embeddings, iteratively refining both. Despite being limited to fixed datasets, some methods convert transduction into induction by reusing the learned similarity function or training a classifier on pseudo-labelled data, enabling generalisation to new inputs. For comprehensive reviews, see Chapelle and Zien [31], Chong et al. [41], Song et al. [206], Subramanya and Talukdar [209].

3.2 Taxonomy

Since the seminal work of Scudder [198], a wide range of SSL methods has been proposed. As discussed earlier, SSL can be divided into inductive and transductive paradigms, with inductive methods further split into generative and discriminative families. Within the inductive–discriminative branch, we categorised methods according to the distributional assumptions they rely on to make the marginal feature distribution $p(x)$ containing information about the conditional distribution $p(y|x)$ (see Section 2.2.2). Figure 5 summarises our taxonomy: at the top, classical non-deep methods are presented, in the middle, their deep extensions are shown, and at the bottom, the adaptation of these methods to medical image segmentation is illustrated.

Classical non-deep methods include mixture models on the generative side, and on the discriminative side, approaches based on the manifold assumption, smoothness and clustering assumptions (like self-training, co-training, boosting), low-density separation (like entropy minimisation or information regularisation), and support vector machines (which is a hybrid method between inductive and transductive paradigms). Transductive SSL is typically represented by graph-based methods that propagate labels across similarity graphs.

The figure also highlights how these foundational ideas extend into the deep learning era. As for non-deep generative SSL, deep generative methods (like GANs, VAEs, diffusion models, energy-based models, and normalising flows) naturally extend to SSL. On the other hand, deep discriminative approaches largely derive from the smoothness and cluster assumptions of classical methods. We group these into consistency-based methods (like Π-model, Mean Teacher, VAT) and entropy-based methods (like FixMatch and related variants), according to the type of unsupervised loss employed (see Section 4.1). Moreover, some methods combine multiple unsupervised losses, for example adding low-density–inspired objectives (as indicated by red arrows in Figure 5). Finally, graph-based methods have also evolved, with modern extensions relying on graph neural networks [206].

The taxonomy also connects the SSL methods for medical image segmentation, where inductive deep discriminative approaches, particularly those based on self-training and co-training, offer the most promising foundations.

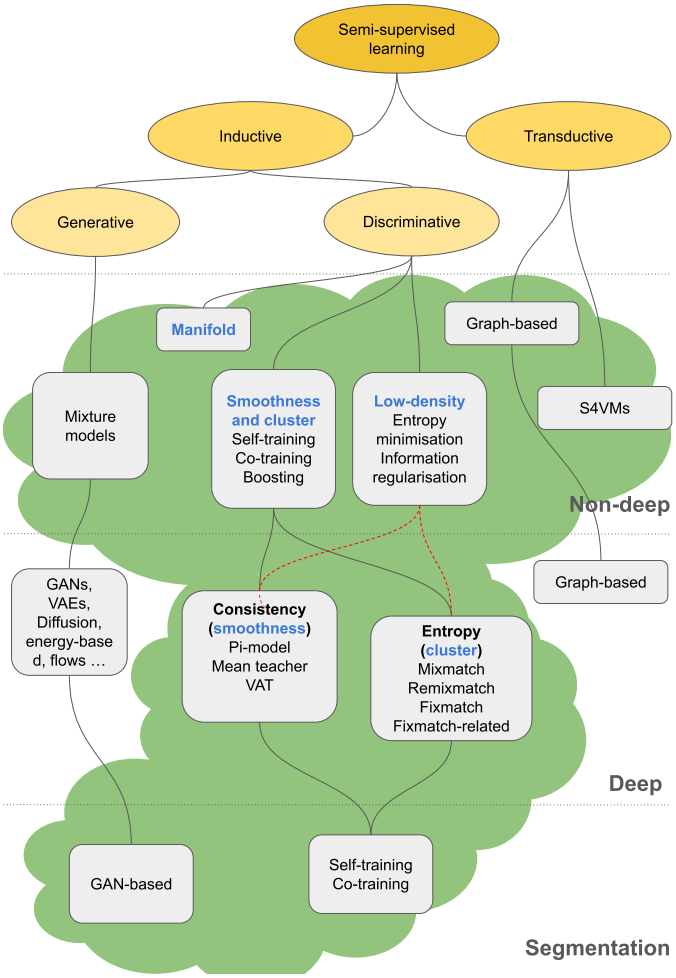

Fig. 5. Our taxonomy of semi-supervised learning methods for classification and segmentation, including data distribution assumptions. All methods discussed in this review are highlighted within the green cloud.

In the previous section, we provided a brief overview of the main non-deep approaches to semi-supervised learning. In the following sections, we focus on deep discriminative SSL methods and their extensions to segmentation. All methods discussed in this review are highlighted within the green cloud on Figure 5.

4 METHODOLOGICAL DEEP SEMI-SUPERVISED METHODS

- méthodes de semipy + extensions
- mettre du code de semipy !
- Parler du problème des dataloaders en SSL (faire un encadré réglé par semipy + faire un truc medium pour que ce soit utilisé !) Parler du pb pour définir des epochs.

4.1 Unlabelled data as a regulariser: classic deep framework

AS: pb de notations à régler unlabeled et labeled (somme de (1)) Deep discriminative semi-supervised learning generally aims to minimise a combination of a supervised loss L and an unsupervised loss H, including unlabelled data. Therefore, the optimisation problem generally becomes finding $\hat{\theta}$ that minimises the following objective:

$$\hat{\mathcal{R}}_{SSL}(\theta) = \frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} L(\theta; x_i, y_i) + \frac{\lambda}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} H(\theta; x_i), \tag{1}$$

where $\mathcal{L}$ and $\mathcal{U}$ denote the sets of labelled and unlabelled indices, respectively, and λ is a scalar weight which balances the labelled and unlabelled loss terms. Alternatively, it is also possible to use the whole dataset for the computation of the unsupervised loss term:

$$\hat{\mathcal{R}}_{SSL}(\theta) = \frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} L(\theta; x_i, y_i) + \frac{\lambda}{|\mathcal{L} \cup \mathcal{U}|} \sum_{i \in \mathcal{L} \cup \mathcal{U}} H(\theta; x_i). \tag{2}$$

For the supervised loss L, most approaches use the cross-entropy loss, defined as

$$L(\theta; x_i, y_i) = -\log p(y_i | x_i; \theta).$$

The choice of the unsupervised loss H is critical and is thoroughly discussed in Schmutz et al. [196], which we briefly summarise below. In most cases, H can be interpreted as a surrogate of the supervised loss L. Depending on how H is defined, deep inductive learning methods can generally be divided into two main categories: **entropy-based methods**, where H is directly an expected form of L, and **consistency-based methods**, where H can be shown to be equivalent to an expected value of L.

For the entropy-based methods, H is typically constructed using the cross-entropy loss, aiming to increase the model's confidence on unlabelled data. A seminal example is entropy minimisation, introduced by Grandvalet and Bengio [85] in a non-deep learning context, but which can be readily adapted to deep discriminative models. In this case, H is defined as the Shannon entropy, and it follows directly that:

$$H(\theta; x) = \mathbb{E}_{\pi_x(\tilde{x}, \tilde{y})} [L(\theta; \tilde{x}, \tilde{y})], \tag{3}$$

with $\pi_x(\tilde{x}, \tilde{y}) = \delta_x(\tilde{x}) p_\theta(\tilde{y}|\tilde{x})$ where δ_x is the Dirac distribution in x. A major subset of entropy-based approaches consists of pseudo-labelling methods, which are discussed in detail in Section 4.1.2. These methods assign a pseudo-label $\hat{y}$ to each unlabeled data point x, defined as the class with the highest predicted probability: $\hat{y} \in \operatorname{argmax}_y p(y|x; \theta)$. To ensure reliability, only pseudo-labels whose associated maximum predicted probability exceeds a predefined confidence threshold τ are retained as targets. This leads to the following unsupervised loss:

$$H(\theta; X) = -\log p(\hat{y}|X : \theta) \mathbb{1}_{\max_Y p(Y|X;\theta) > \tau}$$

In contrast, consistency-based methods aim to enforce prediction invariance under perturbations applied either on the input data or on the model itself. These approaches commonly define H using the Mean Squared Error (MSE). It can be shown that MSE is comparable to cross-entropy in terms of measuring distance between distributions, and we have:

$$C_1 \mathbb{E}_{\pi_x(\tilde{x}, \tilde{y})} [L(\theta; \tilde{x}, \tilde{y})] \leq H(\theta; x) \leq C_2 \mathbb{E}_{\pi_x(\tilde{x}, \tilde{y})} [L(\theta; \tilde{x}, \tilde{y})], \quad \text{with } 0 \leq C_1 \leq C_2. \tag{4}$$

In the following, we review the main deep inductive SSL methods and detail the specific form of H used in each.

4.1.1 Perturbation-based methods. The smoothness assumption suggests that a model's predictions should be stable under small input perturbations, i.e., predictions for clean and slightly noisy inputs should be similar. Since this consistency does not require label information, it can be leveraged using unlabelled data. There are two main ways to enforce this: adding noise to the inputs and penalising prediction differences, or perturbing the model itself. These perturbation-based methods are commonly used in neural networks, where unsupervised loss terms can be naturally integrated, making them well-suited for SSL. In this context, the unsupervised loss term becomes a measure of divergence (D) between the clean output ($f_\theta(x)$) and the perturbed one ($perturbed(f_\theta(x))$):

$$H(\theta; x) = D(f_\theta(x), perturbed(f_\theta(x))). \tag{5}$$

The variations between methods essentially consist in constructing an unsupervised loss by selecting a divergence measure and the type of perturbations to apply (either to the input, either to the model or both). We summarise these choices for the most common SSL methods in Table 2.

Table 2. Perturbation-based methods. The methods in bold are available in the SemiPy library.

Methods	Model's perturbation	Data perturbation	Divergence choice	Notes
Γ-Model	Gaussian noise	None	MSE	
Transformation/Stability	Dropout	Weak	MSE & Mutual exclusivity	
Π-Model	Dropout and noise	Weak	MSE	
Temporal ensembling	Dropout and noise	Weak	MSE	EMA on predictions
Mean Teacher	EMA and dropout	None	MSE	
Dual Student	EMA and dropout	None	MSE	2 teachers, 2 students
Fast-SWA	SWA model averaging	None	MSE	
VAT	Dropout	Adversarial	KL-divergence	Combine with entropy minimisation
VAD	Adversarial dropout	None	KL-divergence or MSE	
ICT	EMA	Mixup	MSE	
MixMatch	EMA	Weak & Mixup	MSE	
ReMixMatch	EMA	Strong, weak & Mixup	Cross-entropy & RotNet loss	Distribution alignment & sharpening
FixMatch & UDA	EMA	Strong & weak	Cross-entropy	

Perturbations on the model: From Γ-Model to Π-Model. First deep discriminative methods in SSL focused on perturbating the model directly. For instance, Rasmus et al. [181] proposed Γ-Model as a simplification of their Ladder Network by removing the decoder. The principle is to pass the inputs (labelled or unlabelled) through the encoder twice: once clean, and once with Gaussian noise added at each layer. The model then minimises the mean squared error (MSE) between clean and corrupted outputs of the encoder's final layer

$$H(\theta; x) = ||f_\theta(x) - \hat{f}_\theta(x)||^2, \tag{6}$$

, where $\hat{f}_\theta$ is the model with the additive Gaussian noise at each layer. Sajjadi et al. [193] proposed a similar idea leveraging stochastic elements of the model like dropout [208]. Additionally, they proposed to also use weak data augmentation and combined an MSE with their earlier mutual-exclusivity loss [192], which encourages confident predictions. Inspired by these works, Laine and Aila [115] introduced the Π-Model using the model's inherent stochasticity of the model through dropout to enforce consistency between two weakly augmented outputs of the same input. The major difference lies in the fact that they apply equal augmentation and noise to the inputs. To stabilise training, the unsupervised weight is gradually ramped up over time. This formulation can also be viewed as a special case of pseudo-ensembles [10], where each input generates two noisy predictions that must agree.

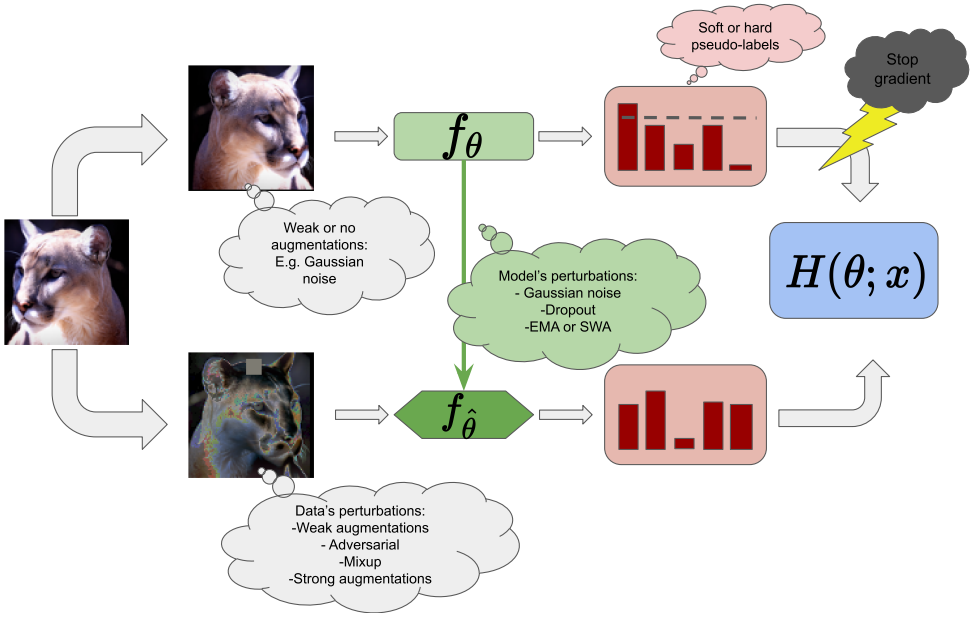

Fig. 6. Perturbations-based models summary.

From stabilising pseudo-labels to encouraging pseudo-labels diversity : Laine and Aila [115] observed that pseudo-labels generated from a single forward pass are unstable due to model training and stochasticity. To address this, they proposed temporal ensembling, which stabilises pseudo-labels by maintaining an exponential moving average (EMA) of model predictions over time:

$$y_{ema}(x) \leftarrow \alpha y_{ema}(x) + (1-\alpha) f_\theta(x), \tag{7}$$

where α is a momentum term that controls the exponential moving average. The unsupervised loss chosen is the MSE, with an unsupervised weight is gradually ramped up over time. While more stable and efficient than $\Pi - Model$, temporal ensembling requires storing historical predictions for all samples. Additionally, pseudo-labels are updated only once per "epoch" for each samples where as the models is updated at every batch. To overcome these issues, Tarvainen and Valpola [217] proposed Mean Teacher. Instead of averaging predictions, they maintain an exponential moving average of model weights to create a teacher model that evolves continuously during training:

$$\theta_{teacher} \leftarrow \alpha \theta_{teacher} + (1-\alpha)\theta, \tag{8}$$

where α is a momentum term that controls the exponential moving average, $\theta_{teacher}$ are the parameters of the teacher model. The teacher generates soft pseudo-labels used to train the student model via an MSE loss. Athiwaratkun et al. [9] further extend this idea using Stochastic Weight Averaging (SWA) [98] instead of EMA, which averages model weights over time under a cyclical learning rate, improving model diversity.

Ke et al. [106] highlight that in the mean teacher framework, the student and teacher become increasingly similar during training, leading to confirmation bias and limited learning. To address this, they introduce two independent student models, each trained separately and exchanging knowledge. Each student is also paired with their own teacher. The total unsupervised loss for a student combines a consistency term with its teacher and a stabilisation loss with

the other student. This dual structure enhances robustness by reducing coupling and confirmation bias but requires training two models.

Adversarial perturbations: Inspired by adversarial training [84, 215], Miyato et al. [153] proposed Virtual Adversarial Training (VAT), a method that computes a virtual adversarial direction (direction in input space that most perturbs the model's output). Unlike earlier approaches that relied on random noise, VAT enforces local smoothness around each input by applying these targeted perturbations, leading to more effective regularisation. This direction, r_{adv}, is found by maximising the KL divergence between the original and perturbed predictions (under a small norm constraint, $||r_{adv}|| \leq \epsilon$). We get an efficient approximation using one gradient step:

$$r_{adv} = \underset{r;||r_{adv}|| \leq \epsilon}{\arg\max} \; KL(f_{\hat{\theta}}(x)||f_\theta(x+r)) \approx \epsilon \frac{\nabla_r D(f_{\hat{\theta}}(x), f_{\hat{\theta}}(x+r))}{||\nabla_r D(f_{\hat{\theta}}(x), f_{\hat{\theta}}(x+r))||}, \tag{9}$$

where $\hat{\theta}$ is a copy of θ with stop-gradient (meaning that the gradient is not propagated through $\hat{\theta}$ during training) and $r \sim \mathcal{N}\left(0, \epsilon/\sqrt{dim(x)} I_{dim(x)}\right)$. The final unsupervised loss uses the KL-divergence, which is equivalent to minimising the cross-entropy between the model's predictions and its perturbed counterpart and maximising the entropy of the model's predictions:

$$H(\theta; x) = KL(f_{\hat{\theta}}(x)||f_\theta(x + r_{adv})) = \mathcal{CE}(f_{\hat{\theta}}(x), f_\theta(x + r_{adv})) - \mathcal{H}(f_{\hat{\theta}}(x)). \tag{10}$$

Combining this with entropy minimisation [85] led to state-of-the-art results at the time.

Park et al. [168] extended this idea to improve perturbations on the model itself with Virtual Adversarial Dropout (VAD) by applying an adversarial dropout mask to the network weights rather than perturbing inputs. The dropout mask is computed following a similar objective as in VAT solved with a 0/1 knapsack problem[107]. The adversarial dropout paper does not specify whether gradients are blocked for $\hat{\theta}$, but as in VAT, they likely are. Later works extended this to channel-wise dropout [121] and Worst-Case Perturbations [262], including both additive and structural perturbations like DropConnect to find the most destabilising changes to the network.

Strong data augmentations: from MixUp to domain-specific. To encourage stronger invariance and improve robustness to adversarial examples, Verma et al. [224] proposed Interpolation Consistency Training (ICT), a method based on the mixup strategy [261] in order to increase the magnitude of the perturbations applied to the inputs. Indeed, Mixup creates interpolated samples by linearly combining two inputs: $Mix_\lambda(a, b) = \lambda a + (1 - \lambda)b$, with $\lambda \sim Beta(\alpha, \alpha)$ with $\alpha \in [0, \infty]$. ICT adapts this to semi-supervised learning by replacing labels with model predictions, leading to the unsupervised loss:

$$D(f_\theta(Mix_\lambda(x, x')), Mix_\lambda(f_\theta(x), f_\theta(x')), \tag{11}$$

where D is typically the MSE. Under the cluster assumption, interpolated inputs likely fall in low-density regions, ideal spots to enforce consistency and push decision boundaries away from class clusters.

Based on this work, Berthelot et al. [17] proposed MixMatch, a holistic SSL method, combining several techniques, notably the Mixup strategy. At each step, the unlabelled batch is transformed into an augmented version following a 4-step procedure: (1) weak augmentations, (2) pseudo-label generations, (3) pseudo-label sharpening to reduce entropy of the predictions, (4) MixUp between elements of the batch. The model is then trained with an MSE on this augmented batch. We note that weak augmentations and MixUp are also applied to the labelled batch. Ablation studies show that the modified Mixup step is critical to MixMatch's strong performance on datasets like CIFAR-10.

A following work, ReMixMatch [16], improves over MixMatch by introducing distribution alignment and augmentation anchoring, and by refining the loss structure. Precisely, distribution alignment encourages the model's predictions on unlabelled data to match the marginal label distribution of the labelled set, $p(y)$. We note that this component inherently implies the MCAR or the MAR assumptions as it assumes that the posterior distribution of the labelled and unlabelled data is the same. Augmentation anchoring states that the pseudo-labels are generated from a weakly augmented version of the inputs and then supervise the predictions on the strong ones. Strong augmentations are generated with CTAugment [16], a combination of data augmentation methods that dynamically tunes augmentation strengths during training. Finally, ReMixMatch also replaces the unsupervised MSE loss from MixMatch with cross-entropy and adds two auxiliary losses: (1) RotNet loss [81] (predicts one of four image rotations applied to each unlabelled sample), (2) CE loss (a cross-entropy term computed on a single strongly augmented version of each unlabelled sample, without Mixup). ReMixMatch outperforms other SSL methods on standard computer vision benchmarks, as confirmed by Wang et al. [230]. However, its complexity, combining many techniques, makes it harder to interpret and less suitable for domains with few labels or where model transparency is critical. Notably, it is not evaluated on NLP or audio tasks due to poor performance when mixing variable-length sequences.

Nevertheless, ReMixMatch showed the SSL community the power of domain specific data augmentations to improve robustness and invariance of the predictions. Therefore, simultaneously, FixMatch [204] and Unsupervised Data Augmentation (UDA) [240] simplified ReMixMatch procedures by only incorporating strong augmentations on unlabelled data, generating pseudo-labels with weak or no augmentations and using a cross-entropy as the unsupervised loss. UDA generates soft pseudo-labels directly from the model's output, while FixMatch uses hard pseudo-labels from weakly augmented inputs, applying a confidence threshold to filter low-confidence predictions. Both methods then train the model using the cross-entropy loss between these pseudo-labels and the predictions on strongly augmented versions of the same inputs. Fixmatch's unlabelled loss term is:

$$H(\theta; x) = \mathbb{1}[\max_y p_{\hat{\theta}}(y|x_1) > \tau] CE(\theta; x_2, \arg\max_y p_{\hat{\theta}}(y|x_1)), \tag{12}$$

where $\hat{\theta}$ is a fixed copy of the current parameters θ indicating that the gradient is not propagated through it, x_1 is a weakly-augmented version of x and x_2 a strongly-augmented one.

Weak augmentations of Fixmatch include standard flips and shifts. For strong augmentations, UDA uses RandAugment [51] for image data and also introduces augmentations for text (e.g., back-translation, TF-IDF word replacement), while FixMatch combines RandAugment, potentially with CTAugment. While Sohn et al. [204] evaluated FixMatch only on computer vision tasks, Wang et al. [230] demonstrated its effectiveness on audio and NLP datasets using domain-specific augmentations such as back-translation for NLP [240] and sub-sampling, gain, pitch, and speed changes for audio. Extensions to tabular data, like VIME [252], adapt FixMatch with tailored augmentations for that domain.

Despite differing in label selection, both Fixmatch and UDA methods achieve comparable performance [230]. By their simplicity, they have inspired many recent SSL approaches. Given their omnipresence in today's SSL literature, we have decided to devote Section 4.1.3 to Fixmatch extensions.

4.1.2 A Focus on Pseudo-Labelling. In deep semi-supervised learning, pseudo-labelling involves generating predictions for unlabelled samples and treating these predictions as labels to compute an unsupervised loss, typically combined with a supervised loss. This idea was introduced in the deep learning context by Lee [119], who proposed using cross-entropy for both labelled data and pseudo-labels. Unlike classical self-training, pseudo-labelling distinguishes between labelled and unlabelled data using a balancing coefficient λ that modulates the contribution of the unsupervised loss. Lee

observed that setting λ too high could disrupt learning even on labelled data. Notably, the method did not include confidence-based filtering, every pseudo-label was treated equally, regardless of quality, making it conceptually close to entropy minimisation [85], which encourages confident predictions on unlabelled samples.

Negative Pseudo-label: In multi-class settings with many possible classes, the chance of assigning the correct pseudo-label to an unlabelled sample diminishes. To address this, Chen et al. [36] introduced negative pseudo-labels in deep SSL setting. Instead of selecting a single likely label, the method selects a small subset of incorrect labels $\mathcal{P}(x)$(elatively small compared to the number of classes) and penalises the model for assigning probability mass to them. If $\mathcal{P}(x)$ is small relative to the total number of classes, it is unlikely to contain the true label. More refined selection of $\mathcal{P}(x)$ can use nearest-neighbour searches or model output probabilities. The corresponding loss is:

$$H(\theta; x) = -\log\left(1 - \sum_{y \in \mathcal{P}(x)} f_\theta(y|x)\right). \tag{13}$$

Combining pseudo-labels and negative pseudo-labels was first explored in Kim et al. [109] and later enhanced by Karaliolios et al. [104] and Duan et al. [64] with FixMatch-based approach and Rizve et al. [187] with pseudo-label selection by addressing uncertainty.

Pseudo-label Filtering: In contrast to Lee's original method, modern approaches typically filter pseudo-labels based on confidence, using a threshold (e.g., $\tau = 0.95$) as in Sohn et al. [204]:

$$\hat{y}_i = \mathbb{1}[f_{\hat{\theta}}(x)_i > \tau], \tag{14}$$

where $0 < i \leq C$, C is the number of classes, and $\hat{\theta}$ indicates no gradient is propagated through the pseudo-label.

Yet, modern neural networks are often poorly calibrated and overconfident [87], which leads to noisy pseudo-labels. To mitigate this, Rizve et al. [187] proposed combining confidence scores with uncertainty estimates. Their approach updates pseudo-labels every T epochs, treating selected pseudo-labels as true labels and incorporating them into the training set. This iterative process avoids the need for an explicit λ but makes the model equally sensitive to both true and pseudo labels, and cannot leverage new training information between updates.

Their method also allows negative and multi-label pseudo-labelling. They combine confidence of the model and uncertainty measure on the prediction (quantified via Monte Carlo dropout [77]) as follows:

$$\hat{y}_i = \mathbb{1}[f_{\hat{\theta}}(x)_i > \tau]\mathbb{1}\left[u\left(f_{\hat{\theta}}(x)_i\right) < \kappa\right], \tag{15}$$

where $u(\cdot)$ is the uncertainty function and κ a threshold. This framework is general and can accommodate various uncertainty estimation methods. Notably, their strong performance also depends heavily on data augmentation. On CIFAR-10 with 1,000 labels, error dropped from 28% without augmentation to 8% with RandAugment [51]; for 4,000 labels, it dropped from 15% to 6%.

In another fashion, Zhou et al. [270] proposed selecting pseudo-labels that remain consistent across training iterations. Interestingly, pseudo-label selection becomes less critical with larger datasets and models. Xie et al. [241] and Yalniz et al. [245] showed that large-scale self-training can tolerate noisy pseudo-labels and even benefit from additional noise (e.g., RandAugment or dropout [208]).

Pseudo-label Generation: Alternative pseudo-label refinement methods include semantic alignment techniques. Jackson and Schulman [99] aligned labelled and unlabelled samples in gradient space to improve pseudo-label consistency.

All Deep Discriminative SSL Methods Are Secretly Pseudo-Label Methods: Most deep discriminative SSL methods can be interpreted as variants of pseudo-labelling. Typically, they generate a prediction and either perturb the input (e.g., Berthelot et al. 16, 17, Miyato et al. 153, Sohn et al. 204, Xie et al. 240) or its prediction (e.g., Laine and Aila 115, Rasmus et al. 181, Tarvainen and Valpola 217). The resulting pseudo-label is used in an unsupervised loss (e.g., cross-entropy, MSE or KL-divergence), making these methods natural extensions of the original pseudo-labeling idea, introduced by Lee [119] in the deep learning context. Some selection processes or refinement techniques of the pseudo-labels are applied differently depending on the method.

Xie et al. [240] further bridged pseudo-labelling with perturbation and graph-based SSL by viewing data points as graph nodes, connected if one can be transformed into the other via perturbation. Assuming the manifold hypothesis, this implies that samples connected through perturbations likely share a label. If each component of the graph contains at least one labelled node, labels can propagate across unlabelled samples. Hence, the strength of the perturbation method (capacity to create a wide diversity of coherent samples) becomes a key factor in SSL performance. With ideal perturbations, methods like UDA or FixMatch could intuitively approach optimal performance.

4.1.3 Extensions of Fixmatch. FixMatch [204] combines pseudo-labelling with strong data augmentation to enforce consistency between weakly and strongly augmented views of the same unlabelled sample. For labelled data, weak augmentations are used, and the model is trained with cross-entropy loss. For unlabelled data, a weakly augmented version generates a hard pseudo-label if its confidence exceeds a fixed threshold τ, which is then used to supervise the strongly augmented version via cross-entropy (see Equation (12)).

FixMatch has garnered significant attention in recent years, with much of the subsequent SSL research focused on extending and improving it, through enhanced data augmentations, better pseudo-label selection, alternative learning strategies, and addressing confirmation bias.

Improving data augmentations: FixMatch's performance strongly depends on effective data augmentation. Several works have aimed to improve its augmentations beyond RandAugment and CTAugment. CutMix [256] and CowMask [75] have been used in place of CutOut[60], improving model performance. Semantic mixing strategies like SciMix [212] and SAMOSA [213] selectively combine semantic and non-semantic parts of different images and have shown benefits when applied to FixMatch and Mean Teacher by improving over MixUp or CutMix.

Adaptive Thresholding Variants. A key limitation of FixMatch is its reliance on a fixed confidence threshold τ for pseudo-label selection (12). Several methods investigate this issue, with the main ones summarised in Table 3. For example, Dash [244] introduces a global threshold τ_t that decays over time and that uses the averaged loss of labelled examples $\hat{\tau}$ to match the model's confidence progression and to overcome the issue of selecting too many incorrect pseudo-labels or eliminating too many correct pseudo-labels. FlexMatch [260] extends FixMatch by learning a threshold $\tau_t(c)$ per class $c \in \{1, \ldots, C\}$. These adapt to each class's learning status based on how frequently it appears in high-confidence predictions for the unlabelled data. Freematch [231] combines the two ideas, by dynamically estimating a global threshold τ_t and adjusting per-class thresholds $\rho_t(c)$ accordingly. More recently, SoftMatch [34] applies per-sample confidence-based weighting to the unsupervised loss, aiming to better balance pseudo-label quality and quantity.

Incorporating Alternative Strategies. Some works combine Fixmatch with alternative strategies, as self-supervised learning. CoMatch [124] combines FixMatch with contrastive learning, entropy minimisation, and graph-based SSL. It constructs graphs of weak and strong augmentation predictions and aligns them via contrastive loss in addition to

Table 3. Adaptive Thresholding Variants of Fixmatch.

Method	Adaptive threshold	Notes
Dash	$\tau_t \propto \gamma^{(t-1)}\hat{\tau},\ \hat{\tau} = \frac{1}{\lvert\mathcal{L}\rvert}\sum_{i\in\mathcal{L}} L(\theta; x_i, y_i),\ \gamma > 1$	optimal value for $\hat{\tau}$ generalisation error bound convergence algorithm
FlexMatch	$\tau_t(c) \propto \sum_{i=1}^{n_u} \mathbb{1}[\max_y p_{\hat{\theta}}(y\lvert x_i) > \tau]\mathbb{1}[\arg\max_y p_{\hat{\theta}}(y\lvert x_i)) = c]$	includes a warm-up phase, generally outperforms FixMatch
Freematch	$\tau_t(c) \propto \rho_t(c)\tau_t$ with τ_t the global threshold and $\rho_t(c)$ the per-class threshold: $\tau_t \propto \lambda_\tau \tau_{t-1} + (1-\lambda_\tau)\frac{1}{n_{u_b}}\sum_{i=1}^{n_{u_b}} \max_y p_{\hat{\theta}}(y\lvert x_i)$ $\rho_t(c) \propto \lambda_\tau \rho_{t-1}(c) + (1-\lambda_\tau)\frac{1}{n_{u_b}}\sum_{i=1}^{n_{u_b}} p_{\hat{\theta}}(y\lvert x_i)\mathbb{1}[\arg\max_y p_{\hat{\theta}}(y\lvert x_i)) = c]$	For t=0: $\tau_t = \rho_t(c) = 1/C$ $\lambda_\tau \in]0,1[$ (EMA) n_{u_b}: # of unlabelled in the batch

the unsupervised loss of FixMatch. This new loss encourages the model to have similar low-dimensional embeddings between similar samples. SimMatch [268] extends CoMatch by directly aligning the distributions of weak and strong embeddings using a distribution-level loss. CRMatch [66] combines FixMatch with RotNet [81], adding a self-supervised rotation prediction loss and an embedding distance loss to improve robustness to strong distortions (strongly distorted images should have the same labels but not the same embedding).

Furthermore, recent methods have integrated negative pseudo-labeled into Fixmatch, to leverage low-confidence samples and learn what labels should not be assigned, improving generalisation [64, 104].

Finally, Xu et al. [243] adapts data programming [182] to SSL by using multiple weak labellers trained on subsets of classes. In DP-SSL, conflicts are resolved through a probabilistic label model to enhance pseudo-label quality.

Tackling Confirmation Bias: Confirmation bias, where the learner struggles to correct its own mistakes when learning from inaccurate pseudo-labels, has been addressed using architectural or optimisation strategies. Chen et al. [33] decouple the pseudo-label generator and learner using two heads, while Wang et al. [229] employ counterfactual reasoning and adaptive margins to prevent label collapse and improve robustness.

Alternative Learning Strategies. There exist also methods that adapt the learning process of Fixmatch. AdaMatch [18] extends it to domain adaptation scenarios. It incorporates logit interpolation, distribution alignment, and adaptive thresholds to address potential shifts between the source and target domains. Notably, AdaMatch uses the same hyperparameters across tasks and does not assume MCAR, making it robust across domains. DeFixMatch [196] addresses estimation bias in semi-supervised risk minimisation. It debiases the unsupervised loss using the labelled data (see Section 4.2.3 for more details), improving theoretical guarantees and calibration without strong distributional assumptions.

4.2 Beyond regularisation

While most of the deep discriminative SSL methods combined a supervised and an unsupervised loss, some other works explored different learning frameworks like bi-level optimisation, combining different learning phases or modifying the classical deep semi-supervised learning framework.

4.2.1 Bi-level optimisation. Bi-level optimisation has emerged as a powerful framework in SSL to address limitations of standard single-objective training. These methods enable more principled control over pseudo-label quality, sample importance, and parameter updates, leading to stronger generalisation under limited labels or non-MCAR situations.

Pseudo-label quality: Meta Pseudo-Labels (MPL; Pham et al. 173) achieved state-of-the-art ImageNet performance from early 2020 to mid-2021[1]. Inspired by Mean Teacher [217] and classic pseudo-labelling [119], MPL trains a student model on both labelled data and teacher-generated pseudo-labels using a cross-entropy loss. A key limitation of conventional pseudo-labelling is that the student can rarely outperform the teacher, since it learns directly from the teacher's outputs. Formally, the student is optimised as

$$\theta_S^* = \arg\min_{\theta_S} \frac{1}{|\mathcal{U}|} \sum_{x \in \mathcal{U}} CE(f_{\theta_T}(x), f_{\theta_S}(x)) := \arg\min_{\theta_S} H(\theta_S), \tag{16}$$

where θ_T and θ_S denote the teacher and student parameters. The goal, however, is to obtain a student that also performs well on the labelled set:

$$L(\theta_S^*) = \frac{1}{|\mathcal{L}|} \sum_{x,y \in \mathcal{L}} CE(y, f_{\theta_S^*}(x)). \tag{17}$$

Because θ_S depends on θ_T, MPL optimises $L(\theta_S^*(\theta_T))$ with respect to θ_T, yielding a bi-level problem:

$$\begin{aligned} \min_{\theta_T} \quad & L(\theta_S^*(\theta_T) \\ \text{s.t.} \quad & \theta_S^* = \arg\min_{\theta_S} H(\theta_S) \end{aligned} \tag{18}$$

This meta-learning setup allows the teacher to adjust its outputs based on feedback from the student, mitigating potential biases. The optimisation is implemented using a two-step gradient update (see Pham et al. 173 for details). Regarding pseudo-label types (soft or hard), the authors found no significant performance difference between hard and soft labels on small-scale datasets (CIFAR-10, SVHN, ImageNet). However, for large-scale experiments with additional unlabelled data (ImageNet with additional external unlabelled images), hard labels were preferred to reduce the computation graph. To further boost performance, MPL incorporates supervised loss on labelled data and the UDA loss [240] on unlabelled data for training the teacher.

Sample importance: As discussed earlier, most deep discriminative SSL methods optimise an objective that combines supervised and unsupervised losses: $\theta^* = \arg\min_\theta L(\theta; \mathcal{L}) + \lambda H(\theta; \mathcal{U})$, where L is typically the cross-entropy on labelled data, H represents the unsupervised loss (e.g., from FixMatch or UDA), and λ is a scalar controlling the influence of the unsupervised term. However, Ren et al. [183] argue that treating all unlabelled data equally may be suboptimal, as pseudo-label quality varies across samples. They propose a per-sample weighting scheme:

$$\theta^* = \arg\min_\theta L(\theta; \mathcal{L}) + \frac{1}{n_u} \sum_{x \in \mathcal{U}} \lambda(x) H(\theta; x), \tag{19}$$

where $\lambda(x)$ is a learned weight specific to each unlabelled sample. This leads to the following bi-level optimisation problem:

$$\begin{aligned} \min_{\lambda} \quad & L(\theta^*, \mathcal{V}) \\ \text{s.t.} \quad & \theta^* = \arg\min_\theta L(\theta; \mathcal{L}) + \frac{1}{n_u} \sum_{x \in \mathcal{U}} \lambda(x) H(\theta; x), \end{aligned} \tag{20}$$

where $\mathcal{V}$ is a small validation set. The problem is solved via two-step stochastic gradient descent. While promising, the effectiveness of this method strongly depends on the size of the validation set, making it challenging to apply in

[1] https://paperswithcode.com/sota/image-classification-on-imagenet

real-world low-label scenarios [183, see Fig. 4, right]. Guo et al. [88] proposed a similar method to handle non-MCAR settings.

Parameters update: In the MNAR setting where unlabelled data may contain unseen classes, He et al. [91] observe that not all network parameters contribute equally to correct classification of the seen classes. Some parameters ("safe") generalise well, whereas others ("harmful") tend to overfit unlabelled data from unseen classes and degrade performance. They formulate a bi-level objective in which one level updates only the safe parameters, while the other suppresses the harmful ones, thereby improving robustness to distribution mismatch.

4.2.2 Multi-phase learning strategies. Recent work has explored multi-phase learning strategies that combine SSL with other forms of incomplete supervision, such as self-supervised learning, or that chain multiple SSL stages. These approaches aim to leverage complementary signals from different training paradigms, improving model performance, particularly in low-label regimes.

Active learning: The idea of combining strategies is not new and started to be explore in the early 2000s in the transductive setting [277]. The combination with active learning (AL) has emerged as an effective strategy to reduce labelling costs while maximising the utility of unlabelled data. Instead of passively relying on random samples, AL actively queries the most informative examples for annotation, complementing SSL's ability to leverage unlabelled data

For instance, Gao et al. [78] proposes a unified framework where an SSL model identifies samples with high consistency loss as candidates for annotation, thereby focusing labelling efforts on points most likely to improve decision boundaries. Similarly, Huang et al. [96] introduces a temporal consistency criterion, selecting samples whose predictions fluctuate across training iterations as indicators of uncertainty.

Self-supervised learning: Although self-supervised learning and SSL share the common goal of learning useful representations from unlabelled data, they rely on different principles. This complementarity allows them to be combined effectively [38, 78, 228, 258, 280]. For example, Chen et al. [38] demonstrate that self-supervised pretraining (e.g., via contrastive learning) followed by semi-supervised fine-tuning can outperform prior state-of-the-art SSL methods. This two-phase pipeline illustrates how strong, general-purpose representations learned through self-supervision can be adapted effectively to semi-supervised tasks, particularly when labelled data is scarce.

A notable example is S4L (Self-Supervised Semi-Supervised Learning) proposed by Zhai et al. [258], which incorporates self-supervised objectives as the unsupervised loss in a semi-supervised framework. S4L explores two auxiliary self-supervised tasks: rotation prediction, where the model classifies the rotation applied to each image, and exemplar learning, where the model learns invariance to transformations [110]. These tasks are integrated with standard SSL techniques, such as VAT, entropy minimisation, and pseudo-labelling, to improve generalisation.

Other approaches pursue joint training strategies, optimising self-supervised and semi-supervised losses simultaneously. For instance, Wang et al. [228] propose a model that jointly optimises a rotation prediction loss and a semi-supervised consistency loss, reinforcing both representation learning and label efficiency. Similarly, Fini et al. [69] introduce a multi-task framework combining a supervised loss on the labelled subset with a self-supervised clustering loss on both labelled and unlabelled data, encouraging clusters to align with the semantic classes in the labelled set.

4.2.3 Unbiased SSL. Schmutz et al. [196] highlighted that the risk minimised by most discriminative SSL methods (see Equation 1) is inherently biased, even asymptotically. This bias impedes the use of standard statistical learning theory

and can hurt empirical performance. To address this, they proposed a simple yet effective correction using the labelled data:

$$\hat{\mathcal{R}}_{DeSSL}(\theta) = \frac{1}{n_l}\sum_{i=1}^{n_l} L(\theta; x_i, y_i) + \frac{\lambda}{n_u}\sum_{i=1}^{n_u} H(\theta; x_i) - \frac{\lambda}{n_l}\sum_{i=1}^{n_l} H(\theta; x_i). \tag{21}$$

This debiased estimator, called DeSSL, is easy to implement and applicable to most of the SSL methods discussed earlier. It provides an unbiased estimate of the true risk $\mathbb{E}_{x,y\sim p(x,y)}[L(\theta; x, y)]$ under the MCAR assumption and enjoys several desirable properties without assumptions on the data distribution: lower variance than the complete-case estimator, proper calibration in terms of scoring rules [82], generalisation error bounds, and consistent parameter estimates with asymptotically normal convergence and reduced variance. Related ideas have been extended to more complex missingness settings: Liu and Goldberg [134] addressed the MAR case, while Sportisse et al. [207] and Hu et al. [95] tackled MNAR scenarios.

In a related direction, Zhu et al. [275] proposed the same objective in a self-training framework, fixing $\lambda = 1$. They argue that this objective defines a doubly robust loss because it remains valid under two extreme scenarios: (1) if the pseudo-labels are completely unreliable, the method effectively falls back to training only on the labelled data when both labelled and unlabelled sample sizes are sufficiently large; and (2) if the pseudo-labels are perfectly accurate, it can fully exploit both the pseudo-labelled and labelled data. Under standard convexity assumptions, they prove that the resulting estimator converges to the true model. They also introduce a MAR extension inspired by Liu and Goldberg [134], though this variant does not enjoy the same theoretical guarantees.

This debiasing idea is also at the origin of predicted-power inference [6, 7]. We propose a review of these methods in Section 6.

5 SAFE METHODS

Safe Semi-Supervised Learning refers to methods that either (1) never empirically degrade performance when unlabelled data is introduced, or (2) come with theoretical guarantees that are equal to or stronger than those of supervised learning on the labelled subset (the complete case baseline). These two perspectives are largely equivalent: stronger theoretical guarantees typically translate into improved performance, such as faster convergence rates or tighter generalisation error bounds; No empirical degradation has to be proven theoretically. In essence, a proof of better performance constitutes a form of safety guarantee.

The literature on safe SSL broadly falls into two categories. The first focuses on providing theoretical guarantees for existing SSL methods, such as co-training, self-training, or pseudo-labelling (see Section 5.1), by analysing conditions or assumptions of efficiency. The second seeks to design new inherently safe algorithms, ensuring that the addition of unlabelled data cannot harm performance under any circumstance (see Section 5.2).

5.1 Theoretical guarantees of existing semi-supervised learning methods

Numerous works have sought to provide theoretical guarantees for common SSL methods. Early efforts focused on co-training, but even today, the success of many SSL techniques remains not fully understood. Most theoretical guarantees depend on strong assumptions about the data distribution or simple cases. A sound theoretical analysis of SSL could clarify the assumptions required, the expected improvements, and its limitations, ultimately guiding practical applications. However, such analysis is often lacking.

Even the most common assumptions used in SSL do not have formal definitions: Wasserman and Lafferty [233] pointed out that smoothness assumptions, though widely invoked, have not been formalised. Rigollet [186] proposed a formal definition of the cluster assumption and studied an SSL clustering method that labels estimated clusters uniformly, yet this definition is absent in much of the literature that references the cluster assumption.

In parallel, numerous works analysed existing methods to provide the necessary conditions to make SSL efficient. For instance, co-training was extensively studied in the early 2000s. Originally proposed by Blum and Mitchell [19], it relied on a multi-view assumption: the features can be split into independent subsets (conditionally independent given the labels), each sufficient for classification. Under these assumptions, Dasgupta et al. [57] showed that co-training achieves better generalisation (a lower generalisation bound) than individual learners. However, the independence assumption rarely holds in practice, especially with limited labels [63]. Balcan et al. [12] relaxed the independence assumption to the expansion assumption (if two subsets are not independent given the labels, they should not be highly correlated), and proved similar theoretical guarantees as Dasgupta et al. [57]. Wang and Zhou [226] showed with a PAC analysis on co-training that the method can even succeed without multi-views, given that the learners have large disagreement. They additionally explained theoretically why co-training does not improve anymore after some iterations and gave an estimation of the number of iterations. In a follow-up work, Wang and Zhou [227] proved a sufficient and necessary condition for co-training to be effective by adopting a graph-view of co-training, stating that the two classifiers should make predictions that are both confident and different on sufficient unlabelled instances, such that one classifier can provide useful additional information to improve the other.

Despite its simplicity, self-training has been analysed repeatedly without reaching a complete theoretical understanding. Culp and Michailidis [54] studied its convergence for specific models like logistic regression. Haffari and Sarkar [89] connected Yarowsky's algorithm with graph-based methods.

Deep SSL methods have received less theoretical scrutiny, often limited to generative models. In the generative setting, Dai et al. [55] analysed the categorical GAN methods and showed the distribution on the input space generated by the generator should be complementary to the true data distribution to further explore the low-density region of the true distribution, i.e., good semi-supervised learning requires a "bad" generator. Athiwaratkun et al. [9] empirically analysed the training trajectories of a stochastic gradient descent over two deep discriminative methods (Π-model and mean teacher) and compared them to the training trajectory of the complete case, showing that their consistency terms lead to flatter loss landscapes near optima, potentially hindering convergence. Zhang et al. [263] derived convergence rates of pseudo-labelling for a two-layer perceptron under strong assumptions (e.g., access to the ground-truth classifier and Gaussian-distributed data). Aminian et al. [5] proposed a new generalisation error bound based on an information-theoretic approach. Recently, Li et al. [123] proposed a theoretical analysis, aimed at answering why FixMatch-like methods generalise better than the complete case under the multi-view data assumption[2] [3] and considering a three-layer CNN. They revealed that FixMatch learns all the discriminative features of each semantic class, while the complete case only randomly captures a subset of features due to the lottery ticket hypothesis [73].

Finally, Mey and Loog [149] provided a comprehensive review of PAC-learning guarantees for SSL under varying strengths of data assumptions, which we recommend for further reading on theoretical connections in SSL.

To summarise, a substantial amount of work has attempted to provide theoretical guarantees for SSL methods, yet even the most common assumptions remain loosely defined or informal. Many analyses introduce new necessary conditions or rely on untestable assumptions about the data distribution, which only hold in highly idealised settings.

[2]This multi-view assumption is not the same as the one used for co-training. Here, the assumption is considered in the sense of Allen-Zhu and Li [3]: each class possesses several discriminative features.

Table 4. Safe semi-supervised learning methods and their definition of safe

Method (by publication year)	**Definition of safe**	
	Theoretical guarantees	**Assumptions**
S4VM [132]	Accuracy no worse than supervised SVMs	Transductive setting, low-density assumption Strong distributional assumptions
DRESS [105]	Model asymptotically normal with reduced variance	$n_l \leq n_u$, No distributional assumptions
Unbiased generative SSL [71]	Unbiased estimator with reduced variance	No distributional assumptions
MCPL [135]	Log-likelihood in the labelled train at least equivalent to the complete case	Gaussian distribution assumption of Linear Discriminant Analysis
UMVP [130]	Improvement over complete case for different measures: top-k precision, F_β score, AUC	Low-density assumption, cluster assumption, Manifold assumption
SSL Sum-Product Networks [220] **(deep)**	Training loss lower than complete case	Sum-Product Networks
Positive Unlabelled SSL [194]	Unbiased risk estimator with reduced variance Generalisation bounds improved with more unlabelled data	Specific loss function Binary classification (extension in [210])
DS^{3}L [88] **(deep)**	Empirical risk on the labelled set no worse than the complete case Generalisation error bounds	No distributional assumption MCAR or non-MCAR settings
DeSSL [196] **(deep)**	Unbiased risk estimator, Lower variance than the complete case Generalisation bounds Asymptotically normal with lower asymptotic variance properties	No distributional assumption MCAR assumption (extension in [95, 207])

Consequently, applying SSL in non-idealised, real-world scenarios requires considerable prior knowledge about the data and can entail significant risks as it lacks theoretical guarantees. For these reasons, the development of safe semi-supervised learning methods is essential for practical applications.

5.2 New safe semi-supervised learning methods

Several research efforts have focused on designing new semi-supervised learning methods that are intrinsically safe, that is, they offer theoretical guarantees that ensure performance equal to or better than the complete case. These methods are summarised in Table 4 for clarity, and presented in the following.

5.2.1 Unbiased risk estimation methods. Sokolovska et al. [205] showed that popular SSL methods like entropy minimisation [85] may lose the asymptotic properties of supervised baselines (because of the unsupervised loss). To address this, they proposed a method that estimates risk using a weighted Monte Carlo estimator, assuming a finite feature space and fully known marginal distribution of x. The weights are computed using both the knowledge of the marginal distribution of x and the whole dataset (labelled and unlabelled). The resulting estimator is asymptotically consistent and normal, with lower asymptotic variance than the complete case baseline. Kawakita and Takeuchi [105] propose a Density Ratio Estimation-based Semi-Supervised estimator (DRESS) by modifying the weights to relax the distributional assumptions, while preserving the theoretical guarantees, although requiring more unlabelled than labelled data.

Li and Zhou [132] introduced safe semi-supervised SVMs, called S4VMs, under the low-density separation assumption. They constructed a pool $\mathcal{M}$ of large-margin separators, y, and defined a gain-loss objective $J(y, y^*, y^{SVM}) = gain(y, y^*, y^{SVM}) - \lambda loss(y, y^*, y^{SVM})$, comparing predictions improvement to the inductive SVM, y^{SVM}, (i.e. complete case). λ is a hyperparameter, *gain* and *loss* denote respectively the gain and the loss in accuracy. Since the ground-truth labels y^* are unknown, they instead optimise a worst-case bound $W(y, y^{SVM}) = \min_{\hat{y} \in \mathcal{M}} gain(y, \hat{y}, y^{SVM}) - \lambda loss(y, \hat{y}, y^{SVM})$ over possible labellings in $\mathcal{M}$. Assuming that the ground truth is realisable and reached by one of the large-margin separators of the pool, this procedure yields a semi-supervised classifier that is guaranteed to perform no

worse than the supervised SVMs. However, these guarantees depend on strong assumptions about the underlying data distribution and if these assumptions are not satisfied, no improvement or degeneration is expected.

Fox-Roberts and Rosten [71] revisited generative SSL and proved new bounds on the bias introduced by unlabelled data in the population likelihood. Their analysis applies to a wide range of generative models and led to the proposal of an unbiased estimator of the population likelihood. This estimator not only corrects the bias caused by missing labels but also exhibits lower variance than the complete case estimator, thus improving convergence. Specifically, they bound the optimal model parameters KL divergence with the KL divergence of the optimal parameters found using biased SSL, highlighting the benefit of their corrected estimator.

Sakai et al. [194] proposed an unbiased risk estimator for binary classification using unlabelled data. The method estimates the risk separately for positive and negative samples, combining them to yield a better global estimate. The authors provided variance guarantees and a generalisation error bound. While initially designed for binary classification, it was extended to multi-class classification in [210, Section 5.7.1].

In Table 4, other methods are presented. Loog [135] proposed a safe method called MCPL, which at least improves the log-likelihood. Li et al. [130] propose safe semi-sUpervised learning for MultiVariate Performance measure (UMVP), proving improvement over the complete case for a chosen measure of performance under common SSL assumptions.

5.2.2 Deep unbiased risk estimation method. Trapp et al. [220] proposed a SSL method for estimating parameters of Sum-Product Networks [177], which are basically acyclic directed graphs with sum and products and can be viewed as deep probabilistic models with clear semantic. They demonstrate lower training loss compared to the complete case.

More recently, Schmutz et al. [196] observe that common SSL losses yield a biased risk estimate, even asymptotically. They propose a very simple bias correction (see Equation 21) that yields an unbiased estimator of the true risk for any deep discriminative SSL methods. They prove that this estimator, called DeSSL (this is also called DeFixmatch when the bias correction is applied to Fixmatch), has lower variance than the complete case, is calibrated under proper scoring rules [82], benefits from generalisation bounds, and leads to consistent, asymptotically normal estimators with lower asymptotic variance. These results require only the MCAR assumption. Similar idea has been explored by Liu and Goldberg [134] for MAR labels (non-deep method), then by Sportisse et al. [207] and Hu et al. [95] for MNAR labels (deep methods).

5.2.3 Deep bi-level optimization method. Guo et al. [88] introduced a bi-level optimisation framework where a neural network, α, learns weights for each unlabelled sample, $\lambda(\alpha; x)$, integrated into the SSL objective. The outer loop minimises the supervised loss after inner-loop optimisation that incorporates the weighted unsupervised loss:

$$\begin{aligned} \min_{\alpha} \quad & \sum_{(x,y)\in\mathcal{L}} L(\theta^*; x, y) \\ \text{s.t.} \quad & \theta^* = \arg\min_{\theta} \sum_{(x,y)\in\mathcal{L}} L(\theta; x, y) + \sum_{x\in\mathcal{U}} \lambda(\alpha; x) H(\theta; x), \end{aligned} \tag{22}$$

Unlike earlier work such as Ren et al. [183], where λ is a scalar, Guo et al. [88] method handles non-MCAR settings, including label distribution shifts, when there are unseen-class unlabelled data. They prove that this method never performs worse than the complete case and provide a generalisation error bound. Related bi-level approaches were proposed by Pham et al. [173] and He et al. [91], but these lack formal safety guarantees under the definition used here.

6 VALIDATION

- mettre des maths
- SSL
- unsupervised
- DA
- PPI

In semi-supervised learning, a major challenge lies in model validation when only a small portion of the data is labelled, or when no labelled data remains for validation. Traditional validation strategies, such as cross-validation, become infeasible or unreliable under such conditions due to the scarcity or absence of labels. This limitation has motivated the development of unsupervised and semi-supervised validation techniques, which aim to estimate model performance or guide model selection without relying heavily on labelled validation sets.

As discussed in Section 2.3.3, realistic evaluation protocols are crucial for the deployment of SSL methods in real-world scenarios. However, current benchmarks often neglect this aspect. For example, in the USB benchmark [230], the test set is used as a validation set for all computer vision tasks, while in audio and natural language processing tasks, the labelled validation sets are even larger than the labelled training sets. Such practices highlight the need for validation strategies that remain effective when labelled data are extremely limited or unavailable.

In this context, a growing line of research focuses on unsupervised or semi-supervised validation, leveraging the abundance of unlabelled data and, when possible, a small labelled subset. Additionally, in extreme scnenarios, no data may remain for validation and one can seek for data-free techniques. Some SSL validation methods even dispense with validation data altogether, aiming instead to assess generalisation indirectly. In this review, validation broadly refers to computing any measure, such as accuracy, cross-entropy, or another proxy, that estimates model performance before test time, typically for purposes like hyperparameter tuning or early stopping. Early unsupervised validation techniques, for instance, were designed to infer the optimal number of clusters [24, 76, 218]. Later, Lange et al. [117] introduced a model assessment framework based on solution stability [22], the principle that a learning algorithm generalises well if its solutions remain consistent across different datasets. Their stability measure upper-bounds the 2-fold cross-validation error in the supervised case and naturally extends to semi-supervised and unsupervised settings.

In this review of validation methods, we also consider two related fields of research: (1) unsupervised domain adaptation (UDA) and (2) prediction-powered inference (PPI). The former focuses on adapting a model trained on a source domain to perform well on a target domain with a different data distribution, thus introducing a non-MCAR perspective on validation, as the distribution shift between domains affects the validity of performance estimation. The latter, PPI, is a statistical framework for combining an experimental dataset with model predictions to perform valid inference. Conceptually, it can be seen as a form of statistical validation, as it quantifies the reliability of model predictions when used to estimate population-level quantities, including validation metrics such as risk or accuracy.

6.1 Unsupervised validation

Donmez et al. [62] introduced an unsupervised approach to estimate the supervised risk for both classification and regression tasks under a maximum likelihood framework. Their method assumes that the marginal distribution of labels, $p(y)$, is known and derives a consistent risk estimator that can be seamlessly integrated with additional labelled

data (see Donmez et al. 62, Section 3). The method is applicable for the independent estimation of a model's risk and model comparison.

Lamiroy and Sun [116] proposed a technique for comparing models using probabilistic interpretations of precision and recall, without requiring labelled data. However, this approach is limited to ranking and relative comparison between models. Similarly, Parisi et al. [167] introduced a ranking method based on the spectral decomposition of the off-diagonal elements of the classifiers' covariance matrix, where the resulting eigenvector entries are proportional to the classifiers' balanced accuracies. Building upon this idea, Jaffe et al. [100] developed a framework to estimate class balance in binary classification, enabling the computation of accuracy through sensitivity and specificity, and later extended their method to the multiclass setting.

Platanios et al. [175] proposed an alternative approach based on inter-model agreement by computing how often models agree on their predictions in a binary classification setting. Assuming independence of classifiers' error, they showed a relation between agreement rates and error rates. More precisely, we can estimate the probability of agreement a of a pair of models, f_{θ_1} and f_{θ_2} using the estimator

$$\hat{a} = \frac{1}{n} \sum_{i \in \mathcal{U}} \mathbb{1}\{f_{\theta_1}(x_i) = f_{\theta_2}(x_i)\}$$

. Assuming the independence of classifiers' error, they prove that:

$$a = p(f_{\theta_1} = f_{\theta_2}) = 1 - p(f_{\theta_1}(X) \neq Y) - p(f_{\theta_2}(X) \neq Y) + 2p(f_{\theta_1}(X) \neq Y \cap f_{\theta_2} \neq Y).$$

Analysing pairs of models, this creates a system of equation solvable as soon as we have independence of the error on at least 3 models. The idea was subsequently extended in Platanios et al. [174, 176] to go beyond the independence assumption.

In a completely different fashion, Deng and Zheng [59] propose an auto-evaluation principle to estimate the accuracy, denoted as a, on an unlabelled test dataset by training a regression neural network to predict model performance:

$$a = A(f_\theta, \mathcal{U}),$$

where A is the auto-evaluation principle that outputs an estimated classifier accuracy. In their paper, A is a linear regressor or a neural network trained with a meta-learning principle highly dependent on data-augmentations. Boyeau et al. [23] also analyse an auto-evaluation principle where models are evaluated on synthetically labelled data with bias correction.

6.2 Semi-supervised validation

Rather than relying on a fully unsupervised validation, Welinder et al. [235] proposed to jointly use labelled and unlabelled data at validation time by estimating the performance curves with confidence bounds. The idea is to model the classifier score with a mixture model in the context of binary classification. Beyond validation, this approach can also assist in classifier recalibration by re-estimating the class-conditional confidence distributions. Similarly, Miller et al. [151] introduced a semi-supervised method for evaluating binary classifiers by analysing their recall characteristics.

Inspired by the co-training paradigm, Madani et al. [143] introduced co-validation, a method for model selection and error estimation in binary classification based on the disagreement between two independently trained models on unlabelled data. The authors demonstrated that per-instance disagreement d provides an unbiased estimate of the error

variance σ^2, serving both as a lower and upper bound on the generalisation error e,

$$\frac{1-\sqrt{1-2d}}{2} \leq e \leq \frac{1+\sqrt{1-2d}}{2}$$

and as an upper bound on the variance of the prediction error across instances $d \geq 2\sigma^2$. Along similar lines, Kääriäinen [103] proposed a semi-supervised generalisation bound also derived from the disagreement probability between pairs of classifiers. More recently, Li et al. [122] developed a semi-supervised approach to estimate model confidence by analysing prediction consistency throughout the training process.

Finally, there exist methods not originally designed for validation but that can be used for this purpose. For instance, Sakai et al. [194] and Schmutz et al. [196] proposed unbiased estimates of the risk using both labelled and unlabelled data, these unbiased estimator can be used on a partially labelled validation set to assess the model's risk. They proved that their estimator is more accurate than the complete case estimator baseline and demonstrated the efficiency of the estimator both theoretically and experimentally (see Section 5.2 for more details on these methods).

6.3 Data-free validation

In certain extreme scenarios, no labelled data may remain for validation, posing a fundamental challenge for model selection and stopping criteria. Nevertheless, several approaches have been proposed to address this setting. Mahsereci et al. [144] introduced an early stopping criterion based solely on local statistics of the training gradients, thereby eliminating the need for a separate validation set. Building on random matrix theory, Feofanov et al. [68] derived a theoretical estimator of classification error in the asymptotic regime that operates entirely without validation data. More recently, Li et al. [125] proposed SLAM, a model selection method that leverages margin distributions and spectral complexity to rank models without relying on labelled validation sets. The authors provide theoretical guarantees bounding SLAM's performance relative to the optimal model and demonstrate its practical effectiveness on FixMatch.

6.4 Beyond MCAR: Unsupervised domain adaptation

Unsupervised domain adaptation refers to the problem of leveraging labelled data in a source domain to learn an accurate model in a target domain where labels are unavailable. Unsupervised domain adaptation can be regarded as a special semi-supervised learning problem. Moreover, it considers that the source and target domains do not have the same distribution, leading to MAR and MNAR settings. While most works focused on learning strategies to get common representations, some methods proposed to use the unlabelled data for model selection.

Some approaches adjust a labelled validation data loss using the density ratio between the marginal distributions of labelled and unlabelled data $p_{\mathcal{U}}(x)/p_{\mathcal{L}}(x)$ to estimate the target risk in a consistent way, such as the importance weighted cross-validation of Sugiyama et al. [211]. You et al. [254] proposed a method that incorporates adapted feature representations into the validation process to achieve an unbiased estimate of the target risk with bounded variance, a variance further reduced using control variates.

Morerio et al. [154] showed that parameters that optimally align the second-order statistics (covariances) of source and target, automatically minimise the entropy of the target domain's predictive distribution under certain assumptions.

Based on the assumption that a well-trained source model should cluster target samples of the same class closely in feature space, Saito et al. [191] measures samples neighbourhood density via entropy of similarity distributions and selects models accordingly. An extension work called MixVal [94] combined this idea with the work of Sugiyama et al. [211].

6.5 Prediction-powered inference

The promise of prediction-powered inference (PPI) lies in its ability to effectively, efficiently, and responsibly leverage machine learning model predictions to support valid statistical inference in scientific applications. Its central objective is to perform reliable inference when an experimental (labelled) dataset is supplemented with model predictions [6, 7, 259]. More precisely, PPI targets a population parameter θ, which may correspond to any statistic of interest computed over the predictions, such as a mean, quantile, regression coefficient, or the minimiser of an M-estimator.

A naive use of model predictions can introduce bias, whereas relying solely on the limited labelled data typically leads to high variance. PPI mitigates both issues by combining predictions from unlabelled data with a bias correction term derived from the labelled subset, yielding valid and statistically efficient inference under the MCAR assumption. This bias-correction principle parallels that of DeSSL [196]. Originally introduced by Angelopoulos et al. [6] and later refined for improved computational and statistical efficiency in Angelopoulos et al. [7], the resulting estimator is given by:

$$\hat{\theta}^{PPI} = \arg\min_{\theta} \frac{1}{n_l} \sum_{i=1}^{n_l} L(\theta; x_i, y_i) + \frac{\lambda}{n_u} \sum_{i=1}^{n_u} L(\theta; x_i, f_\theta(x_i)) - \frac{\lambda}{n_l} \sum_{i=1}^{n_l} L(\theta; x_i, f_\theta(x_i)), \tag{23}$$

which is equivalent to the DeSSL estimator [196] when defining $H(\theta; x) = L(\theta; x, f_\theta(x))$. Consequently, DeSSL's theoretical guarantees, such as asymptotic normality, also apply. PPI thus enables valid inference with tight confidence intervals and well-calibrated p-values [7].

Building on this foundation, several extensions have been proposed. Cross-PPI [282] introduces cross-validation to prevent the waste of labelled samples during training and to improve stability through model averaging. Fisch et al. [70] proposed incorporating stratified sampling over the unlabelled dataset to further reduce the asymptotic variance of the estimator. Miao et al. [150] highlighted that PPI's efficiency strongly depends on prediction quality and introduced a data-adaptive procedure that remains consistent and asymptotically normal, providing valid inference without assumptions on model calibration or accuracy. Finally, Datta and Polson [58] extended PPI beyond the MCAR assumption, addressing the more general MAR setting.

7 APPLICATION OF SEMI-SUPERVISED LEARNING FOR IMAGE SEGMENTATION

Segmentation is the computer vision task of assigning a class label to each pixel in an image, either binary (foreground vs background) or multi-class. However, segmentation is not a classification task at the pixel level, as treating pixels as independent units is an oversimplification. Therefore, the true classification problem, considering all pixel dependencies, would involve a label space of size K^P, where K is the number of classes and P the number of pixels.

A major challenge in (medical or not) image segmentation is imperfect data, in particular: (1) Scarce annotations, due to the high cost and expertise required and (2) Noisy or weak labels, especially for 3D data where exact delineation is difficult. To tackle the first challenge, SSL appears as an appealing solution. However, adapting SSL methods seen in Section 4 is not straightforward. In this section, we review SSL methods for segmentation, including techniques for natural images and focusing on approaches tailored for medical data. But, first, we present a bit of context on the type of model and loss used for segmentation.

Models: Among deep learning models for segmentation, the most prominent are UNet [188] for 2D images and 3DUNet Çiçek et al. [42] or VNet [152] for 3D images. These architectures typically feature an encoder-decoder design with skip connections to preserve spatial resolution. Since, numerous extensions have been proposed. Li et al. [127] propose a hybrid densely connected UNet (H-denseUNet), designed for better performances in 3D contexts, especially

for clinical diagnosis since the radiologist usually observes and segments tumours according three axes. UNet++ [271] redesigned skip connections to aggregate features of varying semantic scales, leading to a more flexible feature fusion scheme than the original UNet. Another drawback of pioneer deep learning models for segmentation is that they do not allow dense inter-patient variation in terms of shape and size in the target organs. Focusing on the pancreas segmentation, Oktay et al. [159]combined UNet with an attention gate focusing on dense target structures and called our final architecture Attention U-Net. Chen et al. [35] proposed TransUNet, a transformer-based variant of UNet, leveraging transformers to provide more powerful encoders. Recently, nnU-Net [97] automatically configures its pipeline using prior knowledge distilled from diverse tasks (achieving state-of-the-art results on 33 out of 53 benchmarks).

Losses: Segmentation performance is typically evaluated using metrics like the Dice score, Jaccard index (Intersection over Union, IoU), and Hausdorff distance [90]. The Dice score is the most widely used, and forms the basis of the popular Dice loss. Many other losses or combinations of losses have been proposed to overcome the specificities of medical images segmentation. Yet, no single loss function outperforms others across all tasks. Ma et al. [141] compared 20 losses and found that compound losses (e.g., Dice + cross-entropy) tend to offer robust performance across varied datasets.

Natural image vs medical image segmentation: PLAN

For a complete list of SSL methods in medical image segmentation, we refer to the curated repository by Luo [136], which includes 262 methods from 2016 to mid-2025[3]. Further surveys can be found in Tajbakhsh et al. [216], Cheplygina et al. [40], and Jiao et al. [101].

7.1 Direct application of SSL methods

Mean teacher, pseudo-label and co-training directly scale to segmentation. But not very efficient as data augmentation super important (cite french)

so how to create diversity: data augmentation ou method design??

method design: (a) cross pseudo supervision, (b) cross confidence consistency (e.g., a component of GCT [17]), (c) mean teacher (used in CutMix-Seg [11]), and (d) PseudoSeg [44] structure (similar to FixMatch [28]).

Perturbation input: domain-adapted Data augmentation: French ... Features perturbations: co-training

7.2 Methods designed for segmentation

7.3 Safe semi-supervised methods for segmentation

7.4 Semi-supervised image segmentation

In between methodological SSL for classification and its application for medical images segmentation lies the application of SSL to semantic segmentation involves assigning a class label to each pixel in a natural image, such as "person" or "car."

Semi-supervised semantic segmentation methods largely mirror trends in SSL classification and fall into three broad categories: generative methods (not discussed here), self-training (input perturbations), and co-training (feature perturbations). While self-training initially struggled to gain traction in semi-supervised semantic segmentation, largely due to challenges in selecting effective domain-specific data augmentations, co-training emerged as the first method to compete successfully with generative approaches (mostly GAN-based).

[3] https://github.com/HiLab-git/SSL4MIS/tree/master

Self-Training: Self-training leverages pseudo-labels derived from confident model predictions, but semantic segmentation introduces unique challenges due to pixel-level correlations and pronounced class imbalance. French et al. [74] analysed the structure of the pixel space and found that low-density regions in the input data do not align well with class boundaries, violating the cluster assumption often used in SSL. To address this, they proposed adapting input perturbations to constrain the orientation of the decision boundary so that it remains parallel to the perturbation directions. While standard augmentations such as cropping, rotation, scaling, and colour jitter preserve the semantic segmentation, they are typically too weak to provide substantial gains over the fully supervised baseline. To enhance the effectiveness of self-training, the authors employed stronger augmentations, specifically CutMix and CutOut, and trained the model within a mean teacher framework. Following this work:

- Olsson et al. [162] introduced ClassMix, mixing predictions from two images by class that improve over CutMix and CutOut.
- Ghiasi et al. [80] simply proposed to copy-paste objects from one image to another.
- Zou et al. [281] used FixMatch/UDA-style framework and improved pseudo-labels with Grad-CAM and attention.
- He et al. [92] tackled class imbalance in pseudo-labels by aligning predicted label distributions with labelled data using adaptive thresholds (to avoir biases toward the majority classes).
- Zhu et al. [278] introduced object-centric cropping to enhance pseudo-label diversity.
- Yang et al. [247] proposed ST++, which avoids iterative training by generating, only once, stable pseudo-labels and selecting reliable unlabelled samples based on consistency across training checkpoints.
- Wang et al. [232] used entropy to filter low-confidence predictions, but incorporated them anyway in a contrastive (InfoNCE) loss, from self-supervised learning.

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Differently, Radosavovic et al. [180]'s data distillation (using internet-scale sources of unlabelled data), aggregating predictions across various augmentations. Zhong et al. [269] and Alonso et al. [4] applied pixel-level contrastive learning to encourage class-consistent representations by pulling similar pixels together and pushing dissimilar ones apart.

Co-training: Co-training emerged as a strong alternative to GAN-based methods:

- Ouali et al. [165] proposed a shared encoder with $K + 1$ decoders (one main trained on labelled data only, K auxiliary). Auxiliary decoders receive perturbed features and are trained with pseudo-labels from the main decoder, ensuring the encoder benefits from unlabelled data.
- Chen et al. [39] introduced cross-pseudo supervision, training two models with different initialisations using each other's predictions as supervision, following exactly the principle of co-training using only one view of the dataset.
- Feng et al. [67] extended cross-pseudo supervision with a dynamically weighted loss to penalise disagreements.
- Fan et al. [65] trained one model conservatively (intersecting predictions) and the other progressively (union of predictions) to diversify learning signals.

Weakly supervised: Some early SSL works consider weakly supervised settings, using a few pixel-level annotations and image-level labels for the rest [120, 166, 234].

7.5 Semi-supervised medical image segmentation: how do the methods scale to real-world datasets?

Like semantic segmentation, SSL in medical image segmentation draws heavily from the SSL classification literature. Interestingly, early work in deep discriminative SSL for medical image segmentation [11] predates similar efforts in semantic segmentation. Medical segmentation tasks may be binary (i.e., pixel-wise classification into foreground or background) or semantic (i.e., multi-class segmentation). As with semantic segmentation, SSL approaches in medical imaging can be grouped into three categories: GAN-based methods, self-training, and co-training. However, unlike semantic segmentation, there is no established set of benchmark datasets. Methods are typically developed for specific clinical tasks and are tested on much smaller, domain-specific datasets.

GAN-based: Zhang et al. [267] introduced an adversarial framework in which a segmentation network and an evaluation network are trained jointly. The evaluator distinguishes between segmentation maps produced from labelled versus unlabelled data. This method was applied to a gland segmentation task using 85 labelled tissue images [203] and 100 additional unlabelled images. Similarly, Nie et al. [156] replaced the evaluator with a confidence network that predicts the trustworthiness of pixel-wise segmentation outputs. Their method was tested on MR images from 50 prostate cancer patients segmented into prostate, bladder, and rectum classes, using 20 additional unlabelled images. Peiris et al. [169] extended this idea by incorporating a dual-view co-training framework: two segmentation models are trained jointly using subsets of the labelled data, with a shared confidence network (termed a "critic"). An agreement loss is imposed on the unlabelled data. This method was tested on three datasets, including one with 670 microscopy images. Similarly, Wu et al. [238] trained two segmentation models collaboratively with two discriminators: one to distinguish label sources (labelled or unlabelled) and another to produce confidence maps, demonstrating performance on a large polyp segmentation dataset.

Self-training: Before the adoption of self-training in semantic segmentation, Bai et al. [11] proposed a self-training method for medical image segmentation. They refined pseudo-labels on unlabelled data using conditional random fields, and tested their method on cardiac MR images with 1,600 slices from 80 patients. Li et al. [128] introduced rotation-equivariant training by minimising consistency loss between predictions on an image and its rotated version, applied to skin lesion segmentation. Bortsova et al. [21] extended this to consistency across two arbitrary transformations, learning to invert the transformation as part of the model. Peng et al. [171] used a similar approach with mutual information regularisation between neighbouring pixels, enforcing prediction consistency between nearby pixels.

Other self-training methods introduced adaptive thresholding for pseudo-labelling. Xu et al. [242] used variational inference to estimate thresholds for distinguishing foreground from background pixels. Perone and Cohen-Adad [172] adapted the mean teacher framework to segment spinal cord grey matter from brain MR images with only 8 labelled subjects out of 20. Cui et al. [52] also employed a mean teacher for brain lesion segmentation in 246 diffusion-weighted images.

Unlike classification, where filtering pseudo-labels by confidence is common, segmentation presents challenges due to spatial dependencies among pixels. Still, several methods address pixel-wise pseudo-label quality. Yu et al. [255] and Sedai et al. [199] used Monte Carlo dropout to select high-confidence pixels for 3D cardiac MR and retinal layer segmentation in optical coherence tomography images, respectively. Cao et al. [26] computed epistemic uncertainty [108] from the variance in predictions across multiple stochastic forward passes (e.g., with dropout and Gaussian noise). Li et al. [129] introduced "self-loop" uncertainty, leveraging a self-supervised auxiliary task to quantify uncertainty at multiple network stages.

As in semantic segmentation, some methods integrate contrastive losses to learn discriminative representations.. For example, Chaitanya et al. [27] and Chaitanya et al. [28] apply pixel-wise contrastive learning to group similar and separate dissimilar pixel embeddings. Luo et al. [139] and Luo et al. [138] proposed a method inspired by ladder networks, using a pyramid prediction network to enforce consistency between multi-scale pseudo-labels and adding uncertainty-based filtering. You et al. [253] combined self-supervised pretraining and semi-supervised training, using a contrastive objective to group similar pixels, inspired by Alonso et al. [4].

Co-training: Co-training is also widely used in medical image segmentation. Peng et al. [170] adapted deep co-training [179] for medical segmentation by generating adversarial examples that encourage disagreement between two models (multiple views), preventing their collapse into similar predictions. The method was later generalised to multi-view co-training. Luo et al. [138] avoided adversarial perturbations by using architectural diversity: one model is a CNN, the other a transformer. Both are trained with co-training to maintain diverse views. Xia et al. [239] used fixed rotations to generate multiple views of the same data and introduced uncertainty-based filtering using epistemic uncertainty [108]. Luo et al. [137] introduced a dual-task setup, where a V-Net backbone supports both a segmentation head and a regression head trained to estimate geometric active contours.

7.6 Safe semi-supervised for image segmentation

Incorporating unlabelled data into a supervised learning framework introduces potential risks, particularly when class imbalance is present or when the MCAR (Missing Completely At Random) assumption does not hold. To address these concerns, several SSL approaches have proposed theoretical guarantees for existing methods, making them "safe" under additional assumptions about the data distribution. Other works have developed new methods that come with theoretical safety guarantees without requiring such assumptions. These are referred to as *safe* semi-supervised learning methods.

To the best of our knowledge, only two works have proposed theoretical safety guarantees specifically for image segmentation. Both focus on medical image segmentation and aim to design frameworks that yield unbiased risk estimators with lower variance than the complete case baseline (i.e., training only on labelled data).

- Wu and Zhuang [237] use a UNet to extract multi-scale features, with the top three levels passed through independent convolutional blocks for segmentation predictions. The predictions from these blocks are concatenated to form the final segmentation map, assuming the MCAR setting. Drawing inspiration from Sakai et al. [194], they propose a positive/negative loss decomposition per class, leading to an unbiased estimator of the risk. Their method demonstrates lower variance in the risk estimate on unlabelled data compared to the complete case baseline.
- Schmutz 195, Chapter 4 propose a second safe SSL method for medical image segmentation. Specifically, they introduce a simple modification of the objective function commonly used by discriminative SSL methods. This approach mirrors the safe classification method proposed by Schmutz et al. [196]. The resulting estimator remains unbiased and is shown to have lower variance than the complete case baseline. Additionally, they provide theoretical guarantees in the form of generalisation error bounds.

8 CONCLUSION, LIMITATION AND PERSPECTIVES

A REGRESSION

In regression tasks, the output space $\mathcal{Y}$ is continuous, in contrast to classification, where $\mathcal{Y}$ is discrete. As a result, the standard SSL assumptions, such as the cluster, manifold, and low-density assumptions, do not naturally extend to regression settings. However, the smoothness assumption, which posits that similar inputs should yield similar outputs, can be readily adapted to regression [233].

Wasserman and Lafferty [233] analysed semi-supervised regression and showed that when the data lies on a low-dimensional manifold, a formulation of the manifold assumption as in Chapelle and Zien [31], this structure can complement the smoothness assumption to yield improved performance over purely supervised methods.

Despite the significant progress in semi-supervised classification and segmentation over the past two decades, semi-supervised regression has received comparatively little attention. For a comprehensive recent survey, see Kostopoulos et al. [111].

Some SSL methods, particularly graph-based transductive approaches, naturally extend to regression. These methods typically propagate information over a similarity graph and estimate continuous values for the unlabeled data as an intermediate step, often prior to applying classification methods [13, 30, 49]. For example, Timilsina et al. [219] introduced a transductive regression method based on heat diffusion with boundary conditions, offering a closed-form solution.

Early inductive approaches, such as self-training and co-training, have also been adapted for regression. Zhou and Li [272] proposed Coreg, a co-training algorithm for regression using two k-NN regressors with distinct distance metrics to capture different data views. In practice, they employed Minkowski distances with different orders. Similarly, Brefeld et al. [25] developed a kernel-based co-training framework for regression that also leverages multiple views.

More recent approaches incorporate deep generative models. Rezagholiradeh and Haidar [184] proposed REG-GAN, a GAN-based method in which the discriminator is trained to both distinguish between real and fake samples and to predict associated continuous values. A related approach by Olmschenk et al. [161] modifies the loss functions to better handle regression objectives within a GAN framework.

Finally, Yin et al. [251] introduced a FixMatch-inspired method for estimating 3D rotations from images. Their approach leverages Fisher-based regressors to jointly predict rotation angles and their associated uncertainties, which are used for filtering pseudo-labels in a regression context.

Safe semi-supervised regression: The question of safeness (see Section 5) when introducing unlabelled data in a supervised learning framework has also been studied in semi-supervised regression. Li et al. [131] proposed a safe method based on co-training and similar to S4VM Li and Zhou [132]. They considered having access to b semi-supervised regressors and one complete case baseline. The idea is to find a convex combination of the b regressors that is never worse than the complete case baseline. Syed and Tahir [214] extended this idea to multi-target regressions. The method of Sakai et al. [194] initially proposed for binary classification and estimating the risk separately for positive and negative samples to combine them afterwards, was extended to ordinal regression [221].

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B REGRESSION

In regression tasks, the output space $\mathcal{Y}$ is continuous, in contrast to classification, where $\mathcal{Y}$ is discrete. As a result, the standard SSL assumptions, such as the cluster, manifold, and low-density assumptions, do not naturally extend to regression settings. However, the smoothness assumption, which posits that similar inputs should yield similar outputs, can be readily adapted to regression [233].

Wasserman and Lafferty [233] analysed semi-supervised regression and showed that when the data lies on a low-dimensional manifold, a formulation of the manifold assumption as in Chapelle and Zien [31], this structure can complement the smoothness assumption to yield improved performance over purely supervised methods.

Despite the significant progress in semi-supervised classification and segmentation over the past two decades, semi-supervised regression has received comparatively little attention. For a comprehensive recent survey, see Kostopoulos et al. [111].

Some SSL methods, particularly graph-based transductive approaches, naturally extend to regression. These methods typically propagate information over a similarity graph and estimate continuous values for the unlabeled data as an intermediate step, often prior to applying classification methods [13, 30, 49]. For example, Timilsina et al. [219] introduced a transductive regression method based on heat diffusion with boundary conditions, offering a closed-form solution.

Early inductive approaches, such as self-training and co-training, have also been adapted for regression. Zhou and Li [272] proposed Coreg, a co-training algorithm for regression using two k-NN regressors with distinct distance metrics to capture different data views. In practice, they employed Minkowski distances with different orders. Similarly, Brefeld et al. [25] developed a kernel-based co-training framework for regression that also leverages multiple views.

More recent approaches incorporate deep generative models. Rezagholiradeh and Haidar [184] proposed REG-GAN, a GAN-based method in which the discriminator is trained to both distinguish between real and fake samples and to predict associated continuous values. A related approach by Olmschenk et al. [161] modifies the loss functions to better handle regression objectives within a GAN framework.

Finally, Yin et al. [251] introduced a FixMatch-inspired method for estimating 3D rotations from images. Their approach leverages Fisher-based regressors to jointly predict rotation angles and their associated uncertainties, which are used for filtering pseudo-labels in a regression context.